

Regulations and Curriculum for
Bachelor of Technology (B.Tech.)
in

Artificial Intelligence and Data Science
Artificial Intelligence and Machine Learning
Computer and Communication Engineering
Computer Science and Engineering
Information Science and Engineering
Robotics and Artificial Intelligence
Computer Science and Engineering (Cyber Security)
Computer Science and Business Systems

Version **2026.01**



(Established under Section 3 of UGC Act, 1956)
Placed under Category 'A' by MHRD, GoI | Accredited with 'A+'
Grade by NAAC

VISION

To build a humane society through excellence in education and healthcare

MISSION

*To develop
Nitte (Deemed to be University)
As a center of excellence imparting quality education,
Generating competent, skilled manpower to face the scientific and social
challenges with a high degree of credibility, integrity,
ethical standards and social concern*

Regulations and Curriculum for

Bachelor of Technology (B. Tech.)

IN

Artificial Intelligence and Data Science
Artificial Intelligence and Machine Learning
Computer and Communication Engineering
Computer Science and Engineering
Information Science and Engineering
Robotics and Artificial Intelligence
Computer Science and Engineering (Cyber Security)
Computer Science and Business Systems

Choice Based Credit System (CBCS)
Effective for the batch admitted during
AY 2026-27

Regulations and Curriculum
B.Tech. Degree Programs
Choice based Credit System
(CBCS)

Effective for the Batch admitted during
Academic Year
2026-27
and onwards

Curriculum for Acquiring Professional Skills (CAPS)
With Scheme of Teaching & Examination

REGULATIONS: 2026 -27

COMMON TO ALL

B.Tech. DEGREE PROGRAMS

CHOICE BASED CREDIT SYSTEM

(CBCS)

Version 2026.01

Choice Based Credit System (CBCS)

1. Choice for the selection of courses during each semester.
2. Choice in planning the academic activities by selecting desired number of courses per semester.
3. Balanced curriculum with engineering, science, humanities, and management courses.
4. Project based learning (PBL), which focuses on experiential learning.
5. Opportunities to study inter-disciplinary courses.
6. Enabling slow learners by offering important courses in all semesters.
7. Optional Summer semester
8. Opportunity to get associated in research projects to acquire research experience.
9. Value addition with Honors / Minor credentials.

Curriculum for Acquiring Professional Skills (CAPS)

1. Practicing outcome-based education (OBE) where courses made student-centric rather than teacher-centric.
2. Provisions for courses integrated with Lab/ PBL component.
3. Focus on Experiential Learning.
4. Ability Enhancement and Skill Development Courses as per National Education Policy (NEP) 2020.
5. Focus on Industry Internship and Research Internship.
6. Students to work on real world/interdisciplinary problems in major project.
7. Importance is given to creativity, innovation, and development of entrepreneurship skills.



Off -Campus Centre, Nitte

&



Off -Campus Centre, Bangalore

Key Information

Program Title	Bachelor of Technology Abbreviated as B.Tech. Artificial Intelligence and Data Science B.Tech. Artificial Intelligence and Machine Learning B.Tech. Computer and Communication Engineering B.Tech. Computer Science and Engineering B.Tech. Information Science and Engineering B.Tech. Robotics and Artificial Intelligence B.Tech. Computer Science and Engineering (Cyber Security) B.Tech. Computer Science and Business Systems
Short Description	Four-year, eight semester Choice Based Credit System (CBCS) type of Undergraduate Engineering Degree Program with English as medium of instruction.
Revision version	2026.01 These regulations may be modified from time to time as mandated by the policies of the University. Revisions are to be recommended by the Board of Studies for ● Artificial Intelligence and Data Science, Artificial Intelligence and Machine Learning, Computer and Communication Engineering, Computer Science and Engineering, Information Science and Engineering, Robotics and Artificial Intelligence, Computer Science and Engineering (Cyber Security), Computer Science and Business Systems ● Humanities ● Mathematics ● Science ● Management and further approved by the Academic Council.
Effective from	01-08-2026
Approvals	● Approved in the 61st Academic Council meeting of NITTE (Deemed to be University), held on 13.06.2025 and vide Notification of Registrar, N(DU) Ref: N(DU)/REG/A2024-25/1395 dated 02-07-2025. ● Approved in the 65th Academic Council meeting of NITTE (Deemed to be University), held on 07.05.2026 and vide Notification of Registrar, N(DU) Ref: N(DU)/REG/AC/2025-26/1096 dated 30.5.2026
Program offered at	1. NMAM Institute of Technology, N(DU) Off -Campus Centre, Nitte, 574110, Karkala Taluk 2. Nitte Meenakshi Institute of Technology N(DU) Off -Campus Centre, Bangalore P.B.No.6429, Yelahanka, Bangalore - 560064
Grievance and dispute resolution	All disputes arising from this set of regulations shall be addressed to the Executive Council. The decision of the Executive Council is final and binding on all parties concerned. Further, any legal disputes arising out of this set of regulations shall be limited to jurisdiction of Courts of Mangalore only.

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PREAMBLE

Nitte Mahalinga Adyanthaya Memorial Institute of Technology (NMAMIT), Nitte, established in 1986 and recognized by the All-India Council for Technical Education, New Delhi, is now a constituent college of Nitte (Deemed to be University), Mangaluru, since June 2022. Rank band 151-200 in the National Institutional Ranking Framework (NIRF) 2024 by the Ministry of Education, Government of India, the College has been placed under 'Platinum' category for having high industry linkages by the AICTE-CII Survey of Industry-Linked Technical Institutes 2020. NMAMIT, the off-campus centre of Nitte DU located at Nitte in Karkala Taluk, has active collaborations with several international universities and organizations for faculty and student exchanges, research, internships and placements.

Nitte Meenakshi Institute of Technology (NMIT), Bangalore, established in 2001 and recognized by the All-India Council for Technical Education, New Delhi, is now a constituent college of Nitte (Deemed to be University), Mangaluru, since October 2024. NMIT is Ranked within the rank band 101-150, in the National Institutional Ranking Framework (NIRF)- 2024 by the Ministry of Education, Government of India. Also, NMIT is accredited with 'A+' Grade by NAAC and Platinum Grade by AICTE-CII Survey of Industry-Linked Technical Institutes 2020. NMIT, the off-campus centre of Nitte Deemed to be University is located at Yelahanka, Bangalore.

NMAMIT:

The Institute offers UG engineering program in fourteen disciplines; Artificial Intelligence & Machine Learning (AIM), Artificial Intelligence & Data Science (AID), Biotechnology (BTY), Civil Engineering(CIV), Computer & Communication Engineering (CCE), Computer Science & Engineering(CSE),Computer Science (Cyber Security)(CYB), Electrical & Electronics Engineering (EEE), Electronics & Communication Engineering (ECE), Electronics (VLSI Design & Technology) (VDT), Electronics & Communication Engineering (ACT), Information Science & Engineering (ISE) , Mechanical Engineering(MEC) and Robotics & Artificial Intelligence (RAI). Out of these programs, seven UG programs i.e., BTY, CIV, CSE, EEE, ECE, ISE and MEC are accredited by NBA, New Delhi under Tier - I category. The institute also offers PG program M.Tech. in five disciplines namely Computer Science & Engineering, Cyber Security, Mechatronics, Structural Engineering and VLSI Design & Embedded Systems as well as MCA program. All the departments have qualified research guides for students interested in taking up research work leading to Ph.D.

NMIT:

The Institute offers UG engineering program in twelve disciplines; Artificial Intelligence & Data Science, Artificial Intelligence & Machine Learning, Aeronautical Engineering(AER), Civil Engineering, Computer Science & Business Systems(CBS), Computer Science & Engineering, Electrical & Electronics Engineering, Electronics & Communication Engineering, Electronics Engineering (VLSI Design & Technology), Information Science & Engineering, Mechanical Engineering and Robotics and Artificial Intelligence. Out of these programs, seven UG programs i.e., AER, CIV, CSE, EEE, ECE, ISE and MEC are accredited by NBA, New Delhi under Tier - I category. The institute also offers PG program M.Tech. in four disciplines namely Electric Vehicle Technology, Structural Engineering, Defence Technology and Robotics & AI as well as MCA and MBA program. All the departments have qualified research guides for students interested in taking up research work leading to Ph.D.

Curriculum: Engineering Programs

The curriculum is jointly approved by members of the Board of Studies (BoS) and Academic Council drawn from academia, Industry, Alumni, and working professionals from Industry, and has been designed to integrate hands-on practical training with the concepts of theory courses to enhance the learning experience.

The Curriculum focuses on students Acquiring Professional Skills (CAPS) through rigorous theoretical training using innovations in pedagogy, experiential learning, active learning, collaborative learning, critical thinking, project planning, Project Based Learning (PBL), Ability enhancement courses for skill-building, effective communication, professional practice, creativity & innovation and developing entrepreneurial skills.

The focus of the Institution is to impart Quality Education to generate competent, Skilled, and Humane Manpower to face emerging Scientific, Technological, Managerial and Social Challenges with Credibility, Integrity, Ethics, and Social Concern.

In the present scenario, students wish to make plans for a bright future. However, student aspirations and industry demands are highly diverse. Employers expect the graduates possess multi-disciplinary competency, Information and Communication Technology (ICT), and leadership skills. In this context, Nitte Deemed to be University offers the opportunity to the students to select the courses of their choice and helps them in grooming to have well-rounded personalities and become industry ready.

Efforts have been made to develop the curriculum, compliant with international professional societies.

As part of providing quality engineering education, Nitte Deemed to be University, has initiated the Choice Based Credit System (CBCS) into its academic curriculum. By this, the students can register for courses of their choice and alter the pace of learning within the broad framework of academic courses and credit requirements. CBCS allows students to plan for their academic load and alter it as they progress in learning. Students also have the option of choosing courses from a pool of courses within each classification. Ample options are given to choose interdisciplinary courses from other programs which will help the student to develop additional skills. Slow learners will also be benefitted since important courses are offered in all semesters. This arrangement helps the students to re-register and clear the backlog courses in the subsequent semester. Suitable provisions are made for fast learners to associate them with research activities of faculty members and contribute to research beyond the working hours.

A mentor helps the student in identifying the courses to be studied in each semester based on program requirements, course prerequisites, student's interest in various disciplines, past academic performance, and courses offered by the departments.

Learning becomes more 'experiential' by carrying out labs associated with theory, mini-projects, and Project Based Learning (PBL) as a part of many courses which enhances the capability of students in understanding and apply Engineering /Technology concepts to solve real life-problems. Hence students will develop the ability to apply the gained knowledge in multi-disciplinary projects and be able to take up major projects based on real-world problems and come up with better solutions while addressing social concerns.

REGULATIONS

COMMON TO ALL B.Tech. (CBCS) DEGREE PROGRAMS OF NITTE (Deemed to be University)

1. INTRODUCTION

- 1.1. The general regulations are common to all B.Tech.(CBCS) Degree Programs offered by NITTE (Deemed to be University) at Nitte and Bangalore Off campuses and shall be called “B.Tech. Regulations”.
- 1.2. The provisions contained in this set of regulations govern the policies and procedures on the Registration of students, imparting instructions of courses, the conduct of the examination & evaluation, certification of student performance, and all amendments related to the said Degree program(s).
- 1.3. This set of Regulations, on approval by the Academic Council and Executive Council, shall supersede all the corresponding earlier sets of regulations of the B. Tech Degree program of NITTE (Deemed to be University) along with all the amendments thereto, and shall be binding on all students undergoing the Graduate Degree Program(s) (Choice Based Credit System) conducted at Nitte and Bangalore Off campuses with effect from its date of approval. This set of Regulations may evolve and get modified or changed through appropriate approvals from the Academic Council / Executive Committee from time to time and shall be binding on all stakeholders. The decisions of the Academic Council (AC) and Executive Council (EC) shall be final and binding.
- 1.4. To guarantee fairness and justice to the parties concerned given the periodic evolutionary refinements, any specific issues or matters of concern shall be addressed separately, by the appropriate authorities, as and when found necessary.
- 1.5. The Academic Council may consider any issues or matters of Concern relating to any or all the academic activities of Engineering courses for appropriate action, irrespective of whether a reference is made here in this set of Regulations or otherwise.
- 1.6. The program shall be called Bachelor of Technology, abbreviated as B.Tech. (Program Title).

2. ELIGIBILITY FOR ADMISSION

Sl. No.	Program	Duration	Eligibility
1	B. Tech.	4 years	Passed 10+2 examination with Physics/ Mathematics / Chemistry/ Computer Science/ Electronics/ Information Technology/ Biology/ Informatics Practices/ Biotechnology/ Technical Vocational subject as per Table-1. Obtained at least 50% marks (40% marks in case of candidates belonging to reserved category) in the above subjects taken together. The specific eligibility requirements will be published in admission brochure from time to time.
2	B.Tech. (Lateral Entry to Second year)	3 years	Passed Minimum THREE years / TWO years (Lateral Entry) Diploma examination with at least 45% marks (40% marks in case of candidates belonging to reserved category) in relevant branch of Engineering and Technology. (The University will offer suitable bridge courses such as Mathematics, Physics, Engineering drawing, etc., for the students coming from diverse backgrounds to prepare Level playing field and desired learning outcomes of the program).

Table-1 Academic Level and Credit Framework for admission to Bachelor of Technology (B.Tech.) degree program			
Sl. No.	Academic Level	Desired Entry Qualifications at different levels.	NHEQF/ NCrF Level at Exit
1	12 th Std.	-	4
2	First Year B.Tech. Degree	12 th Completed (NHEQF level 4 completed)	4.5
3	Second Year B.Tech. Degree	A candidate with a UG Certificate in Engineering or Diploma in the appropriate branch of Engineering/ Equivalent Vocational or Technical Program with NHEQF level 4.5 completed	5

2.1. Qualifications from foreign countries

Candidates with qualifications from educational institutions outside of India may be admitted to the program(s) subject to the establishment of equivalence by the university. The Program Committee will evaluate and establish the eligibility of such candidates.

3. PROGRAM PATHS, EXIT OPTIONS, AND DURATION OF THE B. TECH. PROGRAM

3.1 Program paths, exit options.

Sl. No	Academic Level	Entry Level Qualifications	Qualifications at Exit	NCrF Level
1	1 st yr. of UG Degree	A candidate completing 10+2 years with Diploma of Vocation or passed 12 th std. or equivalent vocational training with NCrF level 4	UG Certificate*	4.5
2	2 nd yr. of UG Degree	A candidate with Diploma in appropriate branch of Engineering/ UG Certificate/ Equivalent Vocational or Technical Program NCrF level 4.5	UG Diploma (Engg.) *	5.0
3	3 rd yr. of UG Degree	A candidate with 10+3+1/12+2/ UG Diploma (Engg.) in appropriate domain with NCrF level 5	B. Sc (Engg.) *	5.5
4	Final yr. of UG Degree	A candidate with 3-year Bachelor's degree B.Sc. (Engg.) with NCrF level 5.5	B. Tech (On completion of 165 credits with a minimum CGPA of 5)	6
	Final yr. of UG Degree with Honours	A candidate with 3-year Bachelor's degree B.Sc. (Engg.) with NCrF level 5.5	B. Tech (Honors) 183 credits (Additional 18 credits over and above 165 credits in the same discipline)	6
	Final yr. of UG Degree with a minor in (Other Discipline).	A candidate with 3-year Bachelor's degree B.Sc. (Engg.) with NCrF level 5.5	B. Tech with Minor 183 credits. Additional 18 credits over and above 165 credits in other disciplines	6

*** It is mandatory to earn 4 credits through Discipline specific Internship/ Training/ Specialised courses before the award of Qualifications at Exit.**

3.2 Duration of the B. Tech. program

- 3.2.1 The B. Tech Program shall extend over a period of a total duration of 4 years for students admitted during the first year of the program.
- 3.2.2 The total duration shall be 3 years for students admitted to the second year under the lateral entry scheme.
- 3.2.3 The maximum period which a student can take to complete a full-time academic program is eight years / Six years for Lateral entry diploma students for B.Tech.
- 3.2.4 Each year shall have the following schedule with 5 ½ days a week. Suggested break down of Academic Year into Semesters.

i)	No. of Semesters / Year	There are three semesters in an academic year. - Two Main semesters (Odd, Even) followed by a summer semester. - Normally the Odd Semester will be from August to December and Even Semester from January to May during a calendar year. - The optional summer semester is offered during the vacation period of the even semester. - The summer semester is offered considering the demand for such courses of needy students, subject to the availability of time, faculty, and other resources under a fast-track mode as the available instructional days during even semester vacation periods are less. However, the number of instructional hours needed to cover the syllabi shall be maintained (equivalent to that in the regular semester) with a greater number of instructional hours per week. Note: The summer semester is primarily to assist slow learners and/or failed students in the main semesters. The summer semester may be used to arrange Add-On courses for other students and/or for deputing them for practical training elsewhere.														
ii)	Semester Duration	Main semester (Odd, Even) each 20 Weeks; Summer Semester 8 Weeks														
iii)	Academic Activities (Weeks)	ODD / EVEN Semester <table><tr><td>Registration of Courses & Course Work</td><td>(16)</td></tr><tr><td>Examination Preparation and Examination</td><td>(04)</td></tr><tr><td>Total</td><td>(20)</td></tr></table> Summer Semester <table><tr><td>Registration of Courses & Course Work</td><td>(05)</td></tr><tr><td>Examination Preparation and Examination</td><td>(03)</td></tr><tr><td>Total</td><td>(08)</td></tr></table> Declaration of results: 02 weeks from the date of the last examination Inter-Semester Recess: <table><tr><td>After each Main Semester</td><td>(02)</td></tr></table> Total Vacation: 10 weeks (for those who do not register for the summer semester) and 4 weeks (for those who register for the summer semester)	Registration of Courses & Course Work	(16)	Examination Preparation and Examination	(04)	Total	(20)	Registration of Courses & Course Work	(05)	Examination Preparation and Examination	(03)	Total	(08)	After each Main Semester	(02)
Registration of Courses & Course Work	(16)															
Examination Preparation and Examination	(04)															
Total	(20)															
Registration of Courses & Course Work	(05)															
Examination Preparation and Examination	(03)															
Total	(08)															
After each Main Semester	(02)															

Note:

- In each semester, there will be provision for students to register for courses at the beginning, dropping of courses in the middle, and withdraw from courses towards the end, under the advice of a faculty member. These facilities are expected to enhance the learning capabilities of students, minimizing their chances of failure in courses registered and ensuring their better monitoring by Faculty Advisors).
- A regular candidate shall be allowed a maximum duration of eight years from the day of admission to Ist semester to become eligible for the award of a Bachelor's degree.
- The calendar of events in respect of the program shall be fixed by the Institution from time to time, but preferably in line with the suggested academic calendar of the NITTE (Deemed to be University).

4. DEGREE PROGRAMS

4.1 Undergraduate B. Tech. Degree Programs are offered in the following disciplines by the respective program hosting departments listed below:

i)	Artificial Intelligence and Data Science	(AID)
ii)	Artificial Intelligence and Machine Learning	(AIM)
iii)	Aeronautical Engineering	(AER)
iv)	Biotechnology	(BTY)
v)	Civil Engineering	(CIV)
vi)	Computer Science & Engineering	(CSE)
vii)	Computer Science & Engineering (Cyber Security)	(CYB)
viii)	Computer and Communication Engineering	(CCE)
ix)	Computer Science and Business Systems	(CBS)
x)	Electrical & Electronics Engineering	(EEE)
xi)	Electronics & Communication Engineering	(ECE)
xii)	Electronics & Communication (Advanced Communication Technology)	(ACT)
xiii)	Electronics Engineering (VLSI Design and Technology)	(VDT)
xiv)	Information Science & Engineering	(ISE)
xv)	Mechanical Engineering	(MEC)
xvi)	Robotics and Artificial Intelligence	(RAI)
Other teaching departments are –		
i)	Chemistry	(CHY)
ii)	Humanities and Social Sciences	(HSS)
iii)	Mathematics	(MAT)
iv)	Management	(MGM)
v)	Physics	(PHY)

4.2 The provisions of these regulations shall apply to any new discipline that may be introduced from time to time and appended to the above list.

5. CREDIT SYSTEM

In the Credit System, the course work of students is unitized, and each unit is assigned one credit after a student completes the teaching-learning process as prescribed for that unit and is successful in its assessment.

5.1 Credit Definition: The following widely accepted definition for credit can provide good flexibility to the students and strengthens CBCS under the University. Here, one unit of course work and its corresponding one credit (while referring to the main semester) shall be equal to:

- Four-credit theory courses (4-0-0) shall be designed for 60 hours of the Teaching-Learning process.
- Four-credit theory + tutorial courses (3-1-0) shall be designed for 45 Lecture hours and 15 tutorial hours of the Teaching-Learning process.
- Three-credit theory courses (3-0-0) shall be designed for 45 hours of the Teaching-Learning process.

- Three-credit theory + tutorial courses (2-1-0) shall be designed for 30 Lecture hours and 15 tutorial hours of the Teaching-Learning process
- Two-credit theory courses shall be designed for 30 hours of the Teaching-Learning process.
- Two-credit theory + Practical courses (1-0-2) shall be designed for 15 Lecture hours and 30 hours of Practical classes of the Teaching-Learning process
- One credit theory course shall be designed for 15 hours of the Teaching-Learning process.

The above figures shall also be applicable in the case of the summer semester. Other student activities which are not demanding intellectually, or which do not lend to effective assessment, like practical training, study tours, and attending guest lectures shall not carry any credit.

5.2 Credit Assignment and Lower & Upper Limits for Course Credits Registration in a Semester

All courses comprise a specific Lecture/Tutorial/Practical/Project (L-T-P-J) schedule. The course credits are fixed based on the following norms.

Lecture / Tutorials / Practical:

- 1-hour Lecture per week is assigned 1.0 Credit.
- 1-hour Tutorial session per week is assigned 1.0 Credit.
- 2-hour Lab. Session/project work per week is assigned 1.0 credit.

For example,

- A theory course with an L-T-P schedule of 3-1-0 hours will be assigned 4.0 credits.
- A laboratory practical course with an L-T-P schedule of 0-0-2 hours will be assigned 1.0 credit.
- Calculation of Contact Hours / Week – A Typical Example

Example:

An L-T-P-C of 2-1-2-4 means 2 instructional units based on classroom lecture (L), one instructional unit of the tutorial (T), and one laboratory (P) based instructional unit all delivered during a calendar week and repeated for the entire duration of the semester to earn 4 credits (C) after passing the course.

- For every academic year, the courses in the curriculum shall be classified as:
 - Semester-Designated Courses (shall be registered only in the semester specified)
 - Year-flexible Courses (may be registered in either the odd or the even semester of the same academic year, subject to availability and student's academic plan).
- As advised by the faculty advisor, a student shall register for a minimum of 18 credits and a maximum of 26 credits in each semester, subject to fulfilment of pre-requisites, wherever applicable; availability of the course during the semester; and approval of the Faculty Mentor.
- Students registering for Minor degree program shall be allowed to register for a maximum of 28 credits from IV semester onwards.
- Students registering for Honors degree program shall be allowed to register for a maximum of 28 credits from V semester onwards.
- The maximum number of credits a student can register during a summer semester shall be 16. However, in special cases, the student may be permitted to register additional 2 credits with the approval of the Department Undergraduate Committee

(DUGC). There is no minimum number of credits fixed for course registration during the summer semester.

6. REGISTRATION

6.1. Every student after consulting his/ her Faculty Advisor in the parent department shall register for the approved courses (core and elective) to earn credits for meeting the requirements of a degree program at the commencement of each Semester on the days fixed for such registration and notified in the academic calendar. Students who fail to register on or before the specified date will be allowed to register within one week of the last date by paying a late fee. Such courses together with their grade and credits earned will be included in the grade card issued by the University at the end of each semester, like ODD, EVEN, and summer and it forms the basis for determining the student's performance in that semester.

- Each course will be identified by a unique Course Code of eight alpha-numerals. The alphabets reflect the discipline to which the course belongs. The number preceding the alphabets indicates the year of implementation. The first numeral (after the alphabet) indicates the learning level (based on prerequisites) of the course, and the rest of the two numerals indicate a running serial number. Example: 25EEE101 represents the course implemented in 2025, offered by EEE Dept., Academic Level-1, course serial number is 01

6.2. Mandatory Pre-Registration for higher semester

To facilitate proper planning of the academic activities of the Semester, the students must declare their intention to register for courses of every semester (2nd Semester and above) at least two weeks before the end of the current semester choosing the courses offered by each department in the next higher semester which is displayed on the Department Notice Board at least 4 weeks before the last working day of the semester. Registration to a higher semester is allowed only if the student fulfills the following conditions.

- Satisfied all the academic requirements to continue with the program of studies.
- Cleared all Institute, hostel, and library dues and fines, if any, of the previous semester.
- Paid all required fees of the Institute and the hostel for the current semester.
- Has not been debarred from registering on any specific grounds by the Institute.

Note:

- (i) Diploma students who are admitted to the second year through Lateral entry should register for mandatory non-credit Mathematics Bridge courses as prescribed, during 3rd and 4th semesters respectively. They shall attend these bridge course classes during the respective semesters to satisfy attendance and CIE requirements.
- (ii) Completion of Mathematics Bridge courses shall be mandatory for the award of the degree.

6.3. Registering for Backlog Courses

6.3.1. Students who have not cleared a course (Theory/ Lab/ project) are shown with

“F” grade. A course having an ‘F’ grade will be considered as a backlog and it has to be re-registered in the subsequent semesters. F-graded courses are eligible to register for the next level course (pre-requisite is met).

6.3.2. Re-registration fee will be as per the university norms existing at the time of re-registration. When a course is re-registered, the evaluation marks of that course shall be treated as canceled/ reset.

6.3.3. To provide an early opportunity for students to clear their backlog of courses, efforts will be made to offer as many courses as possible during Odd, Even and summer semesters.

7. ADD/DROP/AUDIT OPTIONS

7.1 Registration of courses

Each student shall have to register for course work at the beginning of a semester within 2 to 3 days of commencement after discussing with the course teacher and under faculty advice. (Refer Section 5.2)

7.2 DROP-option

During a specified period in the middle of a semester student’s performance in CIE is reviewed by the faculty advisor. Following a poor performance by a student, he/she can be facilitated to drop identified course(s) (up to the minimum credits specified for the semester). Such course(s) will not be mentioned in the Grade card. Such courses are to be re-registered by these students and taken up for study in a subsequent odd / even/ summer semester.

7.3 Withdrawal from courses (Letter Grade “W”)

During a specific period specified towards the end of the semester, a student’s performance in CIE is reviewed by the faculty advisors. Following a poor performance by a student in the identified course (s) he/she is advised to withdraw from such course(s) (up to the minimum credits specified for the semester) with a mention in the Grade card (Grade “W”). Such courses to be re-registered by these students and taken up for study at a later point in time.

7.4 AUDIT-option (Letter Grade “U”)

A student can register for courses for audit only, to supplement his/her knowledge and/or skills. The audit courses shall not be considered in determining the student’s academic performance (SGPA and CGPA) in the semester. “U” grade is awarded to such courses and will be reflected in the grade card on satisfying the attendance requirements and CIE requirements. The candidate need not appear for SEE in such courses. However, CORE courses shall not be made available for audit.

8. PROGRAM STRUCTURE

8.1 Types of courses

A “Course” is defined as a unit of learning that typically lasts one semester, led by one or more teachers, for a fixed roster of students. A course has identified course outcomes, modules/units of study, specified teaching-learning methods, and assessment schemes. A course may be designed to include lectures, tutorials, practical, laboratory work, field work, project work, internship experiences, seminars, self-study components, online learning modules, etc. in any combination.

The following types of courses are included in the B. Tech. program:

- (a) **Humanities, Social Sciences, and Management Courses (HSMC):** These are common courses for all disciplines.
- (b) **Basic Science Courses (BSC):** Physics, Chemistry, and Mathematics: These are mandatory for all disciplines.
- (c) **Engineering Science Courses (ESC):** Basics of Electrical/ Electronics/ Civil/ Mechanical/ Computer Engineering, etc. These are mandatory for all disciplines.
- (d) **Professional Core Courses (PCC):** These are the professional Core Courses, relevant to the chosen specialization/ branch. The core courses shall be compulsorily studied by students, and it is mandatory to complete them to fulfill the requirements of a Program.
- (e) **Professional Elective Courses (PEC):** These are professional Electives, relevant to the chosen specialization/branch and can be chosen from the pool of courses. It shall be supportive to the discipline providing extended scope/enabling exposure to some other discipline /domain and nurturing student proficiency skills.
- (f) **Open Elective Courses (OEC):** These are the Elective Courses from other technical areas and/ or emerging fields. Students of other departments shall opt for these courses for fulfilling the eligibility and prerequisites mentioned in the syllabus.
- (g) **Integrated Professional Core Courses (IPCC):** It refers to Professional Theory Core Course Integrated with Practical of the same course. Credit for IPCC shall be 04 considering L: T: P as 3:0:1 or L: T: P as 2:1:1, (where L, T, and P represent credits not hours per week)
- (h) **Emerging Technology Courses (ETC):** These courses are designed to teach students about developing technologies that will be available within the next five to ten years and are expected to create significant social or economic effects.
- (i) **Programming Language Courses (PLC):** These courses are designed to teach students languages that can be used to communicate with computers for developing and working on different applications.
- (j) **Project Work (PRJ):** Provide experiential learning opportunities for students. Students are required individually, or in a small group, to select and complete a project that may include review, design, development, curation, analysis, etc. with the application of skills and knowledge relevant to the area of study. Project work to be carried out at the parent Institution, or any university / Government recognized organization without affecting the regular class work.
- (k) **Internship (INT):** The internship (a form of experimental learning) program is a workplace-based professional learning experience that offers supervised exposure

to real-life work experience in an area related to the field of study or career interest. An internship may be undertaken at a workplace such as an industry/R&D organization/Government organization, or any other reputed organization/institution recognized for the purpose by the University. The internship program not only helps fresh pass-outs in gaining professional know-how but also benefits corporate sectors. The internship also enhances the employability skills of the student passing out from Technical Institutions.

(l) **Community Engagement (CEN):** enable students to learn about challenges faced by vulnerable households and develop an understanding of local wisdom and lifestyle in a respectful manner. It enables the students to

- Gain an understanding of rural life, Indian culture and ethos and social realities.
- Develop a sense of empathy and bonds of mutuality with the local community.
- Appreciate significant contributions of local communities to Indian society and economy.
- Learn to value the local knowledge and wisdom of the community
- Identify opportunities for contributing to community's socio-economic improvements.

(m) **Mandatory Non-Credit Courses (MNC):** These courses are mandatory, without the benefit of a grade or credit, passing each mandatory course is required to qualify for the award of a degree.

- Assessment of these courses is conducted in the college and will include Continuous Internal Evaluation (CIE). University Semester End Evaluation (SEE) may not be necessary for these courses.
- A minimum of 40% of the prescribed marks of CIE are required to secure a passing grade in these courses.
- The 'PP' grade is awarded for a Pass in the course and the 'NP' grade is awarded for a Fail in the course. In case an 'NP' grade is awarded, the student has to re-register for the same course wherein he has no alternative options.
- The "PP" and "NP" grades do not carry grade points and are hence not included in the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA) computations. However, such non-credit mandatory courses are required to be included in the students' performance records (transcript) with Pass or Fail (PP or NP).
- Courses that come under this category are Employability Skill Development, Kannada etc.

(n) **Ability Enhancement Courses (AEC):** These courses are designed to help students to enhance their skills in language, communication, personality development, etc. They also promote a deeper understanding of courses like social sciences, ethics, culture, human behavior human rights, and the law. Ability Enhancement Courses are based upon the content that leads to knowledge enhancement.

8.2 Typical Breakdown of Courses for the B.Tech. Degree Curriculum:

Sl. No.	Course Category	Credit Range		Suggested Credits
1.	Basic Science Courses (BSC)	30-32		31
2.	Engineering Science Courses (ESC)	12-15	18-21	14
3.	Emerging Technology Courses (ETC)	02		02
4.	Programming Language Courses (PLC)	04		04
5.	Professional Core Courses (PCC)*	52 - 55		55
6.	Professional Elective Courses (PEC)	12-15		12
7.	Open Elective Courses (OEC)	6		06
8.	Humanities, Social Sciences and Management courses (HSMC)	12-15		14
9.	Ability Enhancement Courses (AEC)/Skill Enhancement Courses (SEC)	8-12		11
10.	Mandatory Non-credit Courses (MNC)	Non-Credit		00
13.	Project Work (PRJ)	10		10
14.	Internship (INT)	04		04
15.	Community Engagement (CEN)	02		02
15.	Note: Student can register between 18 to 26 credits per semester			165
	Total minimum Credits to be earned: 165			

* Including One discipline specific vocational Skill enhancement course

8.2.1 The Department Undergraduate Committee (DUGC) will discuss and recommend the exact credits offered for the program for the above components, the semester-wise distribution among them, as well as the syllabi of all undergraduate courses offered by the department from time to time before sending the same to the Board of Studies (BOS). The BOS will consider the proposals from the departments and make recommendations to the Academic Council for consideration and approval.

8.3 The earned Credit Requirements for the B.Tech. Degree is 165.

Degree is awarded by prescribing the total number of credits to be earned, rather than by using the program duration, giving flexibility to a student to plan their career.

8.4 Suggested Program Structure

8.4.1 Suggested distribution of credits for the Common B. Tech Curriculum N(DU) for Nitte and Bangalore off campuses effective from the AY 2026-27

Type of Course		Credits range		Suggested Semester Wise Credit Distribution					Total Credits	
				I & II	III & IV	V & VI	VII	VIII		
Basic Science Courses (BSC)		31		19	12					31
Engineering Science Courses (ESC)		14		14						14
Emerging Technology Courses (ETC)		2				2				2
Programming Language Courses (PLC)		4		4						4
Professional Core Courses (PCC)		52-55			27	20	8			55
Professional Elective Courses (PEC)		12-15				9	3			12
Open Elective Courses (OEC)		6				6				6
Humanities, Social Sciences and Management Courses (HSMC)		14		3	3	5	3			14
Ability & Skill Enhancement Courses (AEC)	Option-1	8		2	4	2				8
	Option-2			4	2					
Foreign Language Course/Research Methodology (AEC)		3					3			3
Community Engagement (CEN)		2						2		2
Professional Practice (PRP)		14	-					14	-	14
Capstone Project/ Research Project (PRJ)		-	10					-	10	14
Internship (INT)		-	4					-	4	
Mandatory Non-credit Courses (MNC)					0	0				0
* Option-1		165		42	46	44	17	16	165	
* Option-2				44	44					

There are two options in choosing Ability Enhancement Courses (AEC)

- NMAMIT Nitte - has chosen Option-1
 - NMIT Bangalore -has chosen Option-2
- (* As applicable)

8.4.2 Suggested Course Offerings During the First Year for Computer Science and allied branches (AIM, AID, CBS, CCE, CSE, CYB, ISE and RAI)

I & II SEMESTER														
Sl. No.	Course Type and Course Code		Course Title	Teaching Department	Contact Hours/Week				SL	Examination				Credits
					L	T	P	J		Duration in Hours	CIE Marks	SEE Marks	Total Marks	
1	BSC	26MAT103	Advanced Calculus	MAT	3	1	-	-	1	3	60	40	100	4
2	BSC	26PHY102	Quantum Computing and Modern Physics	PHY	3	-	2	-	2	3	60	40	100	4
3	BSC	26MAT106	Linear Algebra	MAT	3	1	0	-	1	3	60	40	100	4
4	BSC	26CHY104	Materials Chemistry for Devices and E-Waste Management	CHY	3	0	2	-	2	3	60	40	100	4
5	BSC	26CIV104	Environmental Science & Sustainability	CIV	2	-	-	-	1	3	60	40	100	2
6	BSC	26BTY111	Biology for Engineers	BTY	1	-	-	-	-	1	60	40	100	1
7	ESC	26CSE103	Problem Solving Through Programming	CSE (Allied)	3	-	2	-	2	3	60	40	100	4
8	ESC	26ECE111	Basic Electronics	ECE	3	-	-	-	1	3	60	40	100	3
9	ESC	26ECE101	Applied Digital Logic Design	ECE	2	0	2	-	1	3	60	40	100	3
10	ESC	26EEE102	Elements of Electrical Engineering	EEE	3	0	0		1	3	60	40	100	3
11	ESC	26MEC122	Engineering Visualization	MEC	-	-	2	-	1	-	60	-	60	1
12	HSMC	26HSS102	Universal Human Values & Professional Ethics	Any	2	-	-	-	1	3	60	40	100	2
13	HSMC	26HSS101	Constitution of India & Global Citizenship	HSS	1	0	0	-	-	0	60	0	60	1

14	PLC	26CSE102	Introduction to Python Programming	CSE (Allied) /MEC	2	0	2	-	1	3	60	40	100	3
15	PLC	26MAT107	Mathematics with MATLAB	MAT	0	0	2	-	-	0	60	0	60	1
16	AEC	26HSS131	Communicative English	HSS	1	0	2	-	1	3	60	40	100	2
17	AEC	26HSS132	Knowing Yourself	HSS	-	-	2	-	-		60	-	60	1
18	AEC	26HSS133	Pathways to Success	HSS	-	-	2		-	0	60	0	60	1
Total (Option-1)					32	2	16	-	16	37	960	520	1480	42
Total (Option 2)					32	2	20	-	16	37	1080	520	1600	44

26HSS132 and 26HSS133 are offered only at NMIT, Bangalore

Mandatory Community Engagement							
19	INT	Community Engagement (Activity-based Internship)	1) Social Connect & Responsibility (15 hours lecture to be engaged during second semester)	25 #	0	100	2*
			2) Activity based Internship to focus on the activities pertaining to Social Connect & Responsibility (1 week duration (40 - 45 hours) to be completed during the vacations of II, IV, V & VI Semesters.)	75	0		

To be evaluated for 60 marks and scaled down to 25 marks

* The grades will be included in the VIII semester grade card

8.5 Capstone Project Work/Professional Practice Report

- 8.5.1** Project work can be carried out as domain-specific /interdisciplinary under the guidance of faculty/ faculty members or the students can also opt for Professional Practice in an Industry / Research Institution/Center of excellence.
- 8.5.2** Viva-voce examination shall be conducted individually.

8.6 ELECTIVES

- 8.6.1** A candidate shall take electives in each semester from groups of electives, commencing from the 5th semester.
- 8.6.2** The minimum number of students to be registered for any Elective offered shall not be less than fifteen (15) and should not exceed forty (40).
- 8.6.3** A candidate shall opt for his/her choice of electives and register for the same at the beginning of each of the 5th to 7th semesters if pre-registration is not done. The candidate is permitted to opt for a change of elective within 15 days from the date of commencement of the semester as per the academic calendar of the college.

9. ATTENDANCE REQUIREMENT

- 9.1 Each semester is considered as a unit and the candidate has to put in a minimum attendance of 85% in each subject with a provision of condoning 10% of the attendance by the principal for reasons such as medical grounds, participation in University level sports, cultural activities, seminars, workshops, and paper presentation.
- 9.2 The basis for the calculation of the attendance shall be the term prescribed by the institution by its calendar of events. For the first semester students, the same is reckoned from the date of admission to the course.
- 9.3 The students shall be informed about their attendance position in the first week of every month by the College so that the students shall be cautioned to make up for the shortage.
- 9.4 A candidate having a shortage of attendance (<75%) in any course(s) registered shall not be allowed to appear for SEE of such course(s). Such students will be awarded an 'N' grade in these courses.
- 9.5 He/she shall have to repeat those course(s) with an 'N' grade and shall re-register for the same course(s) core or elective, as the case may be when the particular course is offered next either in a main (odd/even) or summer semester.
- 9.6 Attendance in CIE and SEE: Attendance in all examinations both CIE and SEE of each course registered shall be compulsory and there shall not be any provision for re-examinations. Any student against whom any disciplinary action is pending shall not be permitted to attend any SEE in that semester.

10. WITHDRAWAL FROM THE PROGRAM

10.1 Temporary Withdrawal

A student who has been admitted to a degree program of the college may be permitted once during the program to withdraw temporarily, for one semester, on the grounds of prolonged illness or grave calamity in the family, etc., provided –

- i. The student applies to the College within 6 weeks of the commencement of the college stating fully the reasons for withdrawal together with supporting documents and endorsement from his parent/guardian.
- ii. The College is satisfied with the genuineness of the case and that even by considering the expected period of withdrawal, the student can complete the program requirements (165 credits) within the time limits specified by the university.
- iii. The student does not have any dues or demands at the College / University including tuition and other fees as well as library material.
- iv. A student availing of temporary withdrawal shall be required to pay such fees and/or charges as may be fixed by the college until his/her name appears on the student's roll list. The fees/charges once paid shall not be refunded.
- v. A student will be entitled to avail of the temporary withdrawal facility only once during his/her studentship. However, any other concession for the concerned student shall have to be approved by the academic council.

10.2 Permanent Withdrawal

Any student who withdraws the admission before the closing date of admission for the Academic Session is eligible for the refund of the deposits only. Fees once paid will not be refunded on any account.

Once the admission for the year is closed, the following conditions govern withdrawal of admissions.

- i) A student who wants to leave the College for good will be permitted to do so (and take a Transfer Certificate from the College, if needed), only after clearing all other dues if any.
- ii) Those students who have received any scholarship, stipend, or other forms of assistance from the College shall repay all such amounts.
- iii) The decision of the Principal of the College regarding the withdrawal of a student is final and binding.

11. EVALUATION SYSTEM

11.1 The Academic Performance Evaluation of a student shall be according to a Letter Grading System, based on the Class Performance Distribution.

11.2 The Letter grades O, A+, A, B+, B, C, P, and F indicate the level of academic achievement, assessed on a decimal (0-10) scale.

11.3 The Letter grade awarded to a student in a course, for which he has registered shall be based on his performance in quizzes, tutorials, assignments, etc., as applicable, in addition to two mid-semester examinations and one semester-end examination. The distribution of weightage among these components may be as follows.

Semester End Examination (SEE)		:	Conducted for 100 or 50 marks and further scaled down to 40 marks
Continuous Internal Evaluation (CIE)		:	(60 marks)
CIE for Non-PBL Courses			
i)	Learning activities (LA): Quizzes (MCQs), Tutorials, Assignments, Seminars, Participative learning etc.	:	20 marks
ii)	Sum of two Mid-semester Examinations	:	40 marks
CIE for IPCC Courses or courses with PBL component			
Theory Part: (Assessment =60 marks)			
i)	Learning activities (LA): Quizzes (MCQs), Tutorials, Assignments, Seminars, Participative learning etc.		20 marks
ii)	Sum of two Mid-semester Examinations	:	40 marks
Practical part /PBL component: (Assessment =60 marks)			
i)	Practical part /Project Based Learning (PBL)	:	60 marks
Final; CIE marks for IPCC Courses or courses with PBL component is sum of the marks of Theory Part and Practical part with weightages as given below.			
60% weightage for theory part (0.6x60=36 marks) + 40% weightage for Practical part / PBL component (0.4x60=24 marks)			

Any variation, other than the above distribution, requires the approval of the pertinent DUGC and Academic Council.

11.4 The letter grade awarded to a student in a 0-0-P (Practical) course, is based on an appropriate continuous evaluation scheme that the course instructor shall evolve, with the approval of the pertinent DUGC and the performance in SEE held on the specified period in a semester.

11.5 Evaluation Scheme (Refer to Appendix-B for detailed evaluation guidelines): The course Instructor shall announce in the class and/or display at the Notice board/faculty door/website the details of the Evaluation Scheme, including the distribution of the weightage for each of the components and method of conversion from the raw scores to the letter-grades within the first week of the semester in which the course is offered so that there are no ambiguities in communicating the same to all the students concerned.

11.5.1 Internship: Mandatory Internship is in two parts. Community Engagement (2 weeks-(2 credits)) and Industry Internship (4 weeks-(4 credits))

11.5.1.1 Community Engagement

All the students admitted to the 1st semester of engineering programs shall have to undergo Community Engagement

- Social connect & Responsibility (15 hours lecture to be engaged during second semester)

- Community Engagement to focus on the activities pertaining to Social Connect & Responsibility (1 week duration (40 - 45 hours) to be completed during the vacations, preferably during II Semester or IV semester (only for lateral entry students)
- The internship shall include Inter / Intra Institutional activities. A viva – voce examination (Presentation followed by question-answer session) shall be conducted during the 2nd semester (for lateral entry students, during the 4th semester) and the prescribed credit shall be included in the 8th -semester grade card.
- All the students admitted to the 3rd semester of Engineering programs (Lateral Entry Category) shall have to undergo a mandatory internship of 02 weeks (during the 3rd semester or the intervening period of the 3rd and 4th semesters). The internship shall include Inter/Intra Institutional activities.
- The internship shall be considered as a head of passing and shall be considered for the award of a degree. Those who do not take up / complete the internship shall be declared to fail and shall have to complete it during subsequent University examinations after satisfying the internship requirements. (The faculty coordinator or mentor has to monitor the student's internship progress and interact to guide them for the successful completion of the internship).
- **Procedure for the Evaluation of Community Engagement**
 - Students should submit the reports immediately on completion of the Community Engagement to the respective mentors.
 - The Examination of the Community Engagement will be carried out by the mentor.
 - The Community Engagement shall be slated for 100 marks CIE only and will not have SEE.
 - Community Engagement marks are based on CIE marks (25 marks for the Social Connect and Responsibility, 25 marks for the presentation, and 50 marks for the report and final presentation).
 - A Viva-Voce examination is conducted during II/IV Semesters (Presentation followed by question-answer session) and the prescribed credit shall be included in the VIII semester grade card.

11.5.1.2 Industry Internship

- All the students admitted to engineering programs shall have to undergo Industrial Internship of 04 weeks during the second and third year of their Engineering studies.
- During the intervening period of the IV & V semesters and VI & VII semesters, students shall be ready for industrial experience. Therefore, they shall choose to undergo 4 weeks Internship involving Innovation / Entrepreneurship or at Industry appropriate to the Discipline with the approval of respective HODs.

11.5.2 Capstone Project work/Professional practice evaluation: The evaluation of CIE of the project work shall be based on the progress of the student in the work assigned by the project supervisor, periodically evaluated by him/her together with a department committee constituted for this purpose. Seminar presentation, project report, and final oral examination conducted by the project evaluation committee at the department level shall form the SEE of the project work.

11.5.2.1 In the case of other requirements, such as seminar, field work, or comprehensive viva voce, if any, the assessment shall be made as laid down by the DUGC with the approval of BOS and Academic council.

11.5.2.2 There shall be no re-examination for any course in the credit system.

However, re-registration and examination is permitted for students

- who have abstained from attending CIE or SEE without valid reasons (“N” grade), or
- who have failed (‘F’ grade) to meet the minimum passing standards prescribed for CIE and/or SEE or
- who have been detained for shortage of attendance or who have withdrawn (W grade) who have dropped any course shall be required to re-register for such course(s) and go through CIE and SEE again and obtain a grade equal to or better than “P” Grade in each case.
- While such students should re-register for the same course(s) if core, they can re-register for the alternative course(s) from among the elective courses, as the case may be. The re-registration shall be possible when the particular course is offered again either in a main (Odd/Even) or summer semester.

11.6 Qualifying Standards

Evaluation Method	Qualifying Standard
Continuous Internal Evaluation	Score: $\geq 50\%$ (≥ 30 out of 60 marks)
Semester End Examination	Score: $\geq 40\%$ (≥ 16 out of 40 marks)
For securing a final Pass in a course	Total 50 % of the Course maximum marks (100) to be scored i.e., the sum of the CIE and SEE marks prescribed.

A student shall be declared fail if he/she

- i. Has not satisfied the CIE requirements of any Course(s).
- ii. Has not appeared for the SEE even after satisfying the attendance and CIE requirements.

11.7 Grading System

The letter grade awarded to a student for his/her performance in a course is based on Absolute Grading.

a. Absolute Grading – Letter Grade and its range

The grade point scale for absolute grading

Marks Range (%)	Grade Point	Letter Grade	Descriptor
90 & above	10	O	Outstanding
80-89	9	A+	Excellent
70-79	8	A	Very Good
60-69	7	B+	Good
55-59	6	B	Above Average
50-54	5	C	Average
00-49	0	F	Fails
Absent	0	Ab	Absent

CGPA	Classification
7.00-& above	First Class with Distinction
6.00-6.99	First Class
5.00-5.99	Second Class

- i) **Grade “N”:** A candidate having a shortage of attendance (<75%) in any course(s) or CIE marks less than 50% shall not be allowed to appear for SEE of such course(s). Such students will be awarded an ‘N’ grade in these courses with a grade point of 0.
- ii) The grade points given above help in the evaluation of credit points earned by the student in a course as the credit points are equal to the number of credits assigned to the course multiplied by the grade points awarded to the student in that course. This shall be used in Arriving at the credit index of the student for that semester, as it is the total of all the credit points earned by the student for all the courses registered in that semester.

11.8 Earning of Credits

A student shall be considered to have completed a course successfully and earned the credits if he/she secures an acceptable letter grade in the range of O-P. The letter grade “F” in any course implies the failure of the student in that course and no credits earned.

- a. The Transitional Grades “I”, “W” and “X” would be awarded by the teachers in the following cases. These would be converted into one or the other of the letter grades (O-F) after the student completes the course requirements.
- b. **Grade “I”:** To a student having attendance $\geq 85\%$ and CIE $\geq 70\%$ (35 marks), in a course, but remained absent from SEE for valid & convincing reasons acceptable to the College, like:
 - i. Illness or accident, which disabled him/her from attending SEE.
 - ii. A calamity in the family at the time of SEE required the student to be away from the College.
 - iii. However, the committee chaired by the principal is authorized to relax the requirement of CIE $\geq 70\%$ if the student is hospitalized or advised long-term rest after discharge from the hospital by the Doctor.
 - iv. Students who remain absent for Semester End Examinations due to valid reasons and those who are absent due to health reasons are required to submit the necessary documents along with their request to the Controller of Examinations to write Makeup

Examinations within 2 working days of that examination for which he or she is absent, failing which they will not be given permission.

- c. **Grade “W”:** To a student having satisfactory attendance at classes, but withdrawing from that course before the prescribed date in a semester under Faculty Advice
- d. **Grade “X”:** To a student having attendance $\geq 85\%$ and CIE $\geq 70\%$, in a course but SEE performance could result in an ‘F’ grade in the course. **(No ‘F’ grade will be awarded in this case, but the student’s performance record is maintained separately).**

11.9 Summer / Fast Track semester

- a. The students who have satisfied CIE and Attendance requirements for the course/s and obtained an ‘F’ grade in SEE are permitted to appear directly in ensuing examination/s as backlog paper/s. The students need not re-register for such course/s in the summer / fast track semester. In case the student wishes to improve CIE/ he/she has to re-register for the summer / regular semester as and when offered next.
- b. The student who obtains required attendance and CIE in the summer semester, but obtains an 'F' grade in SEE; is permitted to appear for SEE subsequently as backlog course/s. The student need not repeat the course for Attendance and CIE.
- c. The course/s for which the student does not possess satisfactory attendance and CIE score shall be marked as ‘N’ on the Grade sheet. Such students are not permitted to SEE for the Courses marked as ‘N’ on the Grade sheet. The students have to re-register only for course/s marked as ‘N’ in the summer/ subsequent semester whenever that course is offered and obtain the required CIE and attendance. Subsequently, they are eligible to appear for SEE in such course/s.
- d. Courses with Transitional Grades viz "W", "I", and "X" are also eligible to register in the summer semester in case they wish to improve their score in CIE.
- e. All courses may not be offered in the summer semester. It is the discretion of the University to offer the courses based on the availability of resources. The Institutes shall notify timetable for the summer semester well in advance.
- f. Summer Semester is optional; it is for the student to make the best use of the opportunity.
- g. A student is permitted to register for a maximum of 16 credits in the Summer / fast track semester.
- h. A student has to choose those courses which are offered by the Institution in a given summer Semester.
- i. In the summer semester, each course needs to be offered for the required number of lectures/ tutorial/ laboratory hours as prescribed in the syllabus.

11.10 Grade Card

Each student shall be issued a Grade Card at the end of each semester. This will have a list of all the courses registered by a student in the semester, together with their credits, the letter grades with grade points awarded. Only those courses registered for credit and having grade points shall be included in the computation of the students’ performance like SGPA and CGPA and the courses taken for audit will not form part of this computation. The results of mandatory courses, which are of the non-credit type, shall also be reflected in the Grade card as PP (for Passed) or NP (for not passed). **Each UG student shall have to obtain the grade PP in each mandatory course to qualify for the Degree awarded by the university.**

11.11 Re-evaluation and paper seeing

Re-evaluation is permitted only for theory papers. The University, on receiving application within the stipulated time and remittance of a prescribed fee for re-evaluation, shall permit re-evaluation for the course/s applied. The marks obtained after re-evaluation shall be the final marks awarded.

11.12 The Make-Up Examination

The Make-Up Examination facility would be available to students who may have missed attending the SEE of one or more course(s) in a semester for valid reasons and given the “I” grade; Also, students having the “X” grade shall be eligible to take advantage of this facility. **The makeup examination would be held as per dates notified in the Academic Calendar during the summer semester.** However, it would be possible to hold a makeup examination at any other time in the semester with the permission of the Academic Council of the College. In all these cases, the standard of makeup examinations shall be the same as the regular SEE for the course(s).

- All the “I” and “X” grades awarded to the students would be converted to appropriate letter grades after the make-up examinations. Any outstanding “I” and “X” grades after the last scheduled make-up examinations shall be automatically converted to “F” grades.
- All the “W” grades awarded to the students would be eligible for conversion to the appropriate letter grades only after the concerned students re-register for these courses in a main/ Summer semester and fulfill the passing standards for their CIE and (CIE+SEE).

11.13 Rules for grace marks

Grace marks up to 1% of the maximum total marks of the courses for which he/she is eligible and have registered (non-credit courses excluded) in the examination or 10 marks whichever is less shall be awarded to the failed course(s), (with a restriction of a maximum of 5 marks per course) provided on the award of such grace marks the candidate passes in that course(s)

12. EVALUATION OF PERFORMANCE

The overall performance of a student will be indicated by two indices:

SGPA; which is the Semester Grade Point Average, and CGPA which is the Cumulative Grade Point Average.

SGPA for a semester is computed as follows.

$$SGPA = \frac{\sum [(Course Credits) \times (Grade Point)] (for all courses in that semester)}{\sum [Course Credits]}$$

CGPA is computed as follows:

CGPA

$$= \frac{\sum [(Course Credits) \times (Grade Point)] (for all courses excluding those with F grades until that semester)}{\sum [Course Credits] (for all courses excluding those with F grades until that semester)}$$

13. COMMUNICATION OF GRADES

The SGPA and CGPA respectively, facilitates the declaration of academic performance of a student at the end of a semester and the end of successive semesters. Both would be normally calculated to the second decimal position.

14. REQUIREMENTS FOR VERTICAL PROGRESSION (PROMOTION / ELIGIBILITY TO HIGHER SEMESTERS)

- 14.1. A student must have passed all the courses of I and II semesters to progress from IV semester to V semester.
- 14.2. A student must have passed all the courses of III and IV semesters to progress from VI semester to VII semester.
- 14.3. Lateral entry student must have passed all the courses of III and IV semesters to progress from VI semester to VII semester.

14.4. Termination from the program

A student shall be required to withdraw (discontinue) from the program and leave the college on the following grounds.

- 14.4.1. Failure to secure a minimum CGPA of 5.0 at the end of the 8 years (6 years for lateral entry students).
- 14.4.2. Failure to earn 165 credits (123 /121 as applicable for lateral entry students) in 8 years (6 years for lateral entry students) of duration from the year of admission including the duration of temporary withdrawal (leave of absence).
- 14.4.3. Absence from classes for more than six weeks at a time in a semester without leave of absence being granted by competent authorities.
- 14.4.4. Failure to meet the standards of discipline as prescribed by the college from time to time.

15. AWARD OF CLASS

Sometimes, it would be necessary to provide equivalence of these averages, viz., SGPA and CGPA with the percentages and/or classes awarded as in the conventional system of declaring the results of university examinations. This can be done by prescribing certain specific thresholds in these averages for Distinction, First Class and Second Class. This can be seen in the following Table.

Percentage Equivalence of Grade Points (For a 10-Point Scale)

Grade Point	Percentage of Marks*	Class
≥ 7.00	$\geq 70\%$	First class with Distinction
≥ 6.00	$\geq 60\%$	First Class
$5.0 \leq \text{CGPA} < 6.00$	$50 \leq \text{Percentage} < 60\%$	Second Class

$$\text{Percentage} * = (\text{CGPA}) \times 10$$

16. APPEAL FOR REVIEW OF GRADES

- 16.1. The entire process of evaluation shall be made transparent, and the course instructor shall explain to a student why he/she gets whatever grade he/she is awarded, if and when required. A mechanism for the review of grades is incorporated into the evaluation system. However, before appealing for such review, a student shall first approach the concerned course Instructor and then the concerned DUGC, with the request to do the needful; and only in situations where satisfactory remedial measures have not been taken, the student may then appeal to the Department Academic Appeals Boards (DAAB) before the date specified in Academic Calendar, by paying the prescribed fees.
- 16.2. The fee for such an appeal will be decided by the Senate from time to time. If the appeal is upheld by DAAB, then the fee amount will be refunded to the student.

17. AWARD OF DEGREE

17.1 B.Tech. Degree

- Students shall be declared to have completed the Program of B.Tech. degree and is eligible for the award of degree provided the students have undergone the stipulated Course work of all the semesters under the Scheme of Teaching and Examinations and have earned the prescribed number of credits (165 credits for regular students registered for 4-year degree programs & 123/121 as applicable for lateral entry students).
- For the award of a degree, a CGPA ≥ 5.00 at the end of the Program shall be mandatory.
- Completion of two Bridge courses in Mathematics shall be mandatory for the award of degree to lateral entry diploma students.
- Earning of Activity Points:**
 - As per the AICTE Activity Point Program, every student entering the degree program should earn activity points as shown, for the award of engineering degree.

Sl.No.	Student Category	Activity based points prescribed
1	Regular students admitted to the B. Tech degree program in the first year	100
2	Lateral entry students admitted to the B.Tech degree program in the second year	75
3	Students transferred from other Universities to (i) to 3 rd semester	75
	(ii) to 5 th semester	50

- The activities can be spread over the years (duration of the program) at any time during the semester weekends and holidays, as per the interest & convenience of the students from the year of entry to the program.
- The Activity Points earned shall be reflected on the student's eighth-semester Grade Card.
- Activity Points (non-credit) do not affect SGPA/CGPA and shall not be considered for vertical progression.
- In case students fail to earn the prescribed activity Points before the commencement of 8th-semester examinations, the eighth-semester Grade Card shall be issued only after earning the required activity Points. Students shall be admitted for the award of a degree only after the release of the Eighth semester Grade Card.

17.2 Honours/ Minors Degree**17.2.1 B.Tech. (Honours) Degree**

- i. Students must earn a minimum of 18 additional credits in his/her major program discipline entitles a student to get an 'Honours' credential.
- ii. Students have to pay additional fees for all the courses registered for 'Honours'
- iii. Students with a minimum of 7.5 CGPA and no backlog at the end of the 4th semester will qualify for registering for courses under the 'Honours' credential.
- iv. Students shall register for 'Honours' courses from the 5th semester onwards.
- v. Students should register for additional courses and plan to take courses that are prescribed under that 'Honours' list as per 'pre-requisite' courses to earn the 'Honours' credential.
- vi. Students who wish to acquire an 'Honours' credential need to carry out 'Honours' course registration along with their regular semester course registration.
- vii. He/she accumulates credits by registering for the required courses, and if the requirements for 'Honours' are met within the prescribed minimum time limit of the program, the 'Honours' will be awarded along with the degree.
- viii. Also, the student should meet the following requirements to become eligible for the 'Honours' award.
 - Minimum CGPA of 7.5 in this major discipline at the end of the 8th semester
 - Minimum CGPA of 7.0 in the registered 'Honours' courses
- ix. In case a student withdraws from the 'Honours' registration in the middle of the program, the 'Honours' courses completed will be converted to 'Audit' courses and indicated accordingly in subsequent Grade Sheets and Consolidated Grade Sheets.
- x. It must be noted that the 'Honours' award will be mentioned in the Degree Certificate as "Bachelor of Technology in (specialization) with Honours".
- xi. This fact will also be reflected in the Consolidated Grade Sheet under a separate heading 'Honours' with similar details shown for other credited courses and the CGPA for 'Honours' will be indicated at the end of the list of courses under 'Honours'.
- xii. The grades obtained in the courses credited towards the 'Honours' award is not counted and shall not influence the GPA/ CGPA of the 'program' student has registered.

17.2.2 Minor Degree

- i. Students have to earn a min of 18 additional credits from the courses focused on discipline other than his/her major program discipline entitles a student to get a 'Minor' credential.
- ii. Students have to pay additional fees for all the courses registered for 'Minor'.
- iii. Students with a minimum of 5.0 CGPA and no backlog at the end of the 3rd semester will only qualify for registering for the course under the 'Minor' credential.
- iv. Students shall register for 'Minor' degree courses from the 4th semester onwards.

- v. All Departments will offer 'Minors' in their varied disciplines and will prescribe what set of courses and/or projects is necessary for earning a minor in that discipline.
- vi. Students should register for additional courses and plan to take courses that are prescribed under that 'Minors' list as per 'pre-requisite' courses to earn the 'Minor' credential.
- vii. If any of the courses listed under the 'minor' option is a course listed under his/her curriculum as PCC then the student cannot opt for that 'Minor', since all minor courses need to be earned as additional courses to his/her program curriculum and depts decision is final and binding.
- viii. Students who wish to acquire a 'Minor' can register for 'Minor' courses along with their regular semester course registration.
- ix. Also, the student should have a minimum CGPA of 5.0 in the 'Minor' courses registered to become eligible for the Minor credential. This fact will also be reflected in the Consolidated Grade Sheet under a separate heading 'Minor in (specialization)'.
- x. If the course requirements for a particular 'Minor' are met within the prescribed minimum time limit of the program, the minor will be awarded along with the degree, and it will be mentioned in the Degree Certificate as "Bachelor of Technology in (Major discipline) with Minor in (specialization)."
- xi. In case a student withdraws from the 'Minor', the 'Minor' courses completed, will be converted to 'Audit' courses and indicated accordingly in subsequent Grade Sheets and Consolidated Grade Sheets.
- xii. The grades obtained in the courses credited towards the 'Minor' award are not counted and shall not influence the GPA/ CGPA of the program the student has registered for.

17.2.3 Additional norms for Honours/Minors

- i. Students shall register for additional courses to earn Honours/Minors in consultation with their Class Advisor from the list of courses suggested by the DUGC.
- ii. DUGC may recommend Massive Open Online Courses (MOOCs)/SWAYAM/NPTEL courses to students who wish to register for Honours/Minors after justifying and establishing the equivalence of the curriculum. The decision of DUGC should be communicated to the Dean of Academics and Controller of Examinations for seeking approval.
- iii. The credits prescribed for Honors/Minors may be earned through SWAYAM/NPTEL
- iv. Students may choose to take up additional course work, from the MOOCs courses list suggested by various departments (which can be from SWAYAM/NPTEL) with proctored examinations as approved by the University and complete the same before the last working day of the VIII semester with a final score (online assignments: 25 % + Proctored examination: 75 %) leading to the following certificates: Completed the course (40-59)– ELITE (60 to 75 %) or ELITE + SILVER (76 to 89 %) or ELITE + GOLD (≥ 90 %)

- v. In case, there is no proctored examination for SWAYAM/NPTEL Courses, the University will conduct a SEE as deemed to be fit for the award of Credits
- vi. The Credit equivalence for online courses shall be as follows –
 - 4 weeks of online course duration – 1 credit (approx. 13-14 hours)
 - 8 weeks of online course duration – 2 credits (approx. 26-28 hours) and
 - 12 weeks of online course duration – 3 credits (approx. 39-42 Hours)

17.2.4 Examination for Swayam Courses

- a) Students are advised to check well in advance the availability of SWAYAM courses to be enrolled in the SWAYAM portal (**Ref: Notification of Registrar N(DU)/REG/AC/2024-25/1040**)
- b) Students must complete the entire SWAYAM course and submit a minimum of 75% of assignments and quizzes to be eligible for the end-term examination
- c) End-term proctored examinations for all SWAYAM-based credit courses are conducted by the National Testing Agency (NTA) and National Programme on Technology Enhanced Learning (NPTEL) at their designated centers across the country.
- d) Alternatively, for select courses, the university will conduct end-term examinations during the current semester in alignment with the regular academic calendar. These exams will be conducted in accordance with the UGC's framework to facilitate learning on SWAYAM.
- e) The end-term examination will carry 70-75% weightage, while continuous assessments consisting of assignments and quizzes completed on the SWAYAM portal will carry 25-30% weightage depending on the evaluation plan announced by SWAYAM course coordinators.
- f) Completion certificate from SWAYAM must be submitted within the first week of commencement of respective semesters.
- g) Re-examinations will be conducted in the subsequent two semesters for students who could not pass or appear in the end-term exams.

17.3 Noncompliance

17.3.1 Noncompliance with Capstone Project/Professional Practice

The Capstone Project/Professional Practice shall be considered as a head of passing and shall be considered for the award of degree. Those, who do not take up/complete the Capstone Project/Professional Practice shall have to complete the same during subsequent University examinations after satisfying the Capstone Project/Professional Practice requirements.

17.3.2 Noncompliance of Internship

All the students of B. Tech shall have to undergo mandatory Internship-for 4 weeks to earn 4 credits-during the vacations at the end of the 2/3 academic year. The evaluation of Internship shall be during VIII semester. The internship shall be considered mandatory for the award of a degree. Those, who do not take up/complete the internship shall have to complete the same during subsequent University examinations after satisfying the internship requirements.

The maximum duration for a student for complying with the Degree requirements is 16 – semesters from the date of first registration for his/ her first semester (8 years from the date of admission to the first year, (12 semesters / 6 years from the date of admission for lateral entry student)).

18. GRADUATION REQUIREMENTS AND CONVOCATION

- 18.1 A student shall be declared to be eligible for the award of the degree if he/she has
- a) Fulfilled “Award of Degree” Requirements
 - b) No Dues to the College, Departments, Hostels, Library, Central Computer Centre and any other centers
 - c) No disciplinary action is pending against him/her.
- 18.2 Students shall be admitted for the award of the degree only after the release of the Eighth Semester Grade Card.
- 18.3 The award of the degree must be recommended by the Executive Council.
- 18.4 **Convocation:** Degree will be awarded to the students who have graduated during the preceding academic year. Students are required to apply for the Convocation along with the prescribed fees, after having satisfactorily completed all the degree requirements (refer to “Award of Degree”) within the specified date to arrange for the award of the degree during convocation.

19. AWARD OF PRIZES, MEDALS, CLASS & RANKS

- 19.1 For the award of Prizes and Medals, the conditions stipulated by the Donor may be considered as per the statutes framed by the University for such awards. Sometimes, it would be necessary to provide equivalence of these averages, viz., SGPA and CGPA with the percentages and/or Class awarded as in the conventional system of declaring the results of university examinations. This can be done by prescribing certain specific thresholds in these averages for Distinction, First Class, and Second Class as described in Section 15.
- 19.2 An attempt means the appearance/registration of a candidate for an examination in one or more courses either in part or failing a particular examination.
- a. A candidate who fails/remains absent (after submitting exam application) in the main examination and passes one or more subjects/courses or all subjects/courses in the summer / Make-up examination shall be considered as taken more than an attempt.
- 19.3 Merit Certificates and University Medals/ will be awarded based on overall CGPA, governed by the specific selection criteria that may be formulated by the University for such Medals / Awards
- a. Only those candidates who have completed the Program and fulfilled all the requirements in the minimum number of years prescribed (i.e., 3 years for Diploma lateral entry students or 4 years for students who joined after the 12th standard) and who have passed each semester in the **first attempt** are eligible for the award of Merit Certificates and /or University Medals.

- b. Candidates with W, N, I, X & F grades and who passes the courses in the subsequent **summer** /make up examinations are not eligible for the award of Gold Medal or Merit Certificate.

20. CONDUCT AND DISCIPLINE

- 20.1** Students shall conduct themselves within and outside the premises of the College in a manner befitting the students of an Institution of National Importance.
- 20.2** **As per the order of the Honorable Supreme Court of India, ragging in any form is considered a criminal offense and is banned. Any form of ragging will be severely dealt with.**
- 20.3** The following acts of omission/ or commission shall constitute a gross violation of the Code of Conduct and are liable to invoke disciplinary measures:
- i. Ragging.
 - ii. Lack of courtesy and decorum; indecent behavior anywhere within or outside the campus.
 - iii. Willful damage or stealthy removal of any property/belongings of the College/Hostel or fellow students/citizens.
 - iv. Possession, consumption, or distribution of alcoholic drinks or any kind of hallucinogenic drugs.
 - v. Mutilation or unauthorized possession of Library books.
 - vi. Noisy and unseemly behavior, disturbing studies of fellow students.
 - vii. Hacking in computer systems (such as entering into another Person's area without prior permission, manipulation and/or Damage of computer hardware and software, or any other Cybercrime, etc.).
 - viii. Plagiarism of any nature.
 - ix. Any other act of gross indiscipline as decided by the Senate from time to time.
 - x. Use of Mobile in the college Academic area.
 - xi. Smoking in College Campus and use of any Tobacco products.
 - xii. Unauthorized fundraising and promoting sales.
 - xiii. Commensurate with the gravity of the offense the punishment may be: reprimand, expulsion from the hostel, debarring from an examination, disallowing the use of certain facilities of the College, rustication for a specified period or even outright expulsion from the College, or even handing over the case to appropriate law enforcement authorities or the judiciary, as required by the circumstances.
- 20.4** For an offense committed in (i) a hostel (ii) a department or a classroom and (iii) elsewhere, the Chief Warden, the Head of the Department, and the Dean (Academics), respectively, shall have the authority to reprimand or impose fine.
- 20.5** All cases involving punishment other than reprimand shall be reported to the Head of the Institution.
- 20.6** Cases of adoption of unfair means and/or any malpractice in an examination shall be reported to the Controller of Examinations for taking appropriate action.
- 20.7** **Note:** Students are required to be inside the examination hall 20 minutes before the commencement of the examination. This is applicable for all examinations (Semester end **summer** /makeup) henceforth. Students will not be allowed inside the examination hall after the commencement, under any circumstances.

21. Pursuing Two Academic Programs

- 21.1. Nitte (Deemed to be University) permits students to pursue two academic programs simultaneously, in alignment with the National Education Policy (NEP) 2020 and the UGC's guidelines (Ref: Notification of Registrar N(DU)/REG/AC/2024-25/1039)
- 21.2. Regulations and guidelines from statutory regulatory authorities will take precedence over these regulations, any updates or amendments to the regulatory guidelines will be incorporated into the university's regulations as required.
- 21.3. The university reserves the right to amend these regulations as required.

21.3.1. Eligibility

- Students are permitted to pursue a second academic program of the same qualification level – i.e., undergraduate students can enrol for a second undergraduate level program and post-graduate students can enrol for another post-graduate level program.
- Students may enrol for the second program at Nitte (Deemed to be University) or any other recognized university.
- PhD students are not permitted to pursue any other academic program along with their PhD.

21.3.2. Modes of Learning:

- The second program may be pursued in-person, online, or through Open and Distance Learning (ODL) mode.
- Students must seek prior written approval from Nitte (Deemed to be University) before enrolling in the second academic program.
- The application for approval must include details of the second program and its alignment with the student's primary program of study.
- A student can pursue two full time academic programmes in physical mode provided that in such cases, class timings for one programme do not overlap with the class timings of the other program.
- Degree or diploma programmes under ODL/Online mode shall be pursued with only such HEIs which are recognized by UGC/Statutory Council/Govt. of India for running such programmes.

APPENDIX - A

Definitions, terminology, and abbreviations

1. Nitte DU / University

- a. Refers to Nitte (Deemed to be University)

2. EC

- a. Refers to Executive Council of Nitte (Deemed to be University)

3. BoS

- a. Refers to the Board of Studies in appropriate Branches of Engineering

4. Institute/Institution

- a. Refers to NMAM Institute of Technology, Nitte / Nitte Meenakshi Institute of Technology, Bangalore

5. Program

- a. A range of learning experiences over a specified period, leading to the award of a degree/diploma/certificate. A program is completed when the courses that make up the program are completed, and other requirements as specified in the program regulations are met.

6. Course

- a. A unit of learning that typically lasts one semester, led by one or more teachers, for a fixed roster of students. Often referred to as a “subject”. A course has identified course outcomes, modules/units of study, specified teaching-learning methods, and assessment schemes. A course may be designed to include lectures, tutorials, practical, laboratory work, fieldwork, project work, internship experiences, seminars, self-study components, online learning modules, etc. in any combination.

7. Semester

- a. An academic session, usually of 16 weeks duration, with a minimum of 90 working days during which coursework and assessments are to be completed. Typically, two semesters make up an academic year, with the first of these referred to as the Odd Semester and the second as the Even Semester.
- b. An additional short semester (usually 8 weeks) may be offered between an even semester and subsequent odd semester (in the interval between two academic years) and is termed a summer semester. The summer semester is offered to enable students to register for:
 - i. Fast-tracked courses required for clearing backlog courses.
 - ii. Fast-tracked courses for earning additional credit / completing non-credit mandatory requirements.
 - iii. Value added courses.
 - iv. The courses offered in summer semesters are bound by the same regulations as that of regular semesters, except that they are run at an accelerated pace to provide the required contact hours and conduct assessments within the 8 weeks.

8. Credit

- a. A unit by which the course work is measured. It determines the number of hours of formal learning (contact hours) required per week. Credits are calculated based on the concept of “notional learning time”. Notional learning time is the number of hours that a learner is expected to spend, on average, to achieve the specified learning outcomes of the course. This may comprise a variable combination of scheduled learning activities, (lectures, seminars, labs, etc.) and supervised learning time (reading required before classes, working on assignments, examination preparation, and completion of assessments).

9. Credit equivalence of notional learning time for different types of activities

- a. The credit values assigned to various teaching-learning activities are as follows:

Type of teaching-learning	Nature of activity	No. of contact hours per week equivalent to one credit	The total number of contact hours over a 16-week semester is equivalent to one credit
Lectures / Seminars / synchronous virtual classes / synchronous webinars	Scheduled instruction	1:1	15
Tutorials	Scheduled instruction	2:1	30
Supervised Demonstrations / Laboratory sessions / Studio / Workshops / Workplace simulation / Skill Practice Sessions	Scheduled instruction	2:1	30
Supervised Field visits/community visits/Internships	Scheduled instruction	3:1	45
Scheduled supervised study (individual or group)	Scheduled instruction	2:1	30
Asynchronous E-Learning modules (structured self-directed study)	Independent learning	2:1	30
Student Seminar	Independent /small group learning	2:1	30
Project work/dissertation	Independent /small group learning	3:1	45
Internship for credit	Industry placement/ Research Internship	3:1	40-45

10. Choice-based credit system (CBCS)

A program structure for higher education requires students to earn a minimum of credits by completing various types of courses, including electives, which facilitate a student to have some freedom in selecting his/her own choices, within as well as across disciplines.

11. Course Registration

Refers to formal registration of the Courses in the study every semester (Credits and Audit) by every student under the supervision of a faculty advisor. The institution will maintain records of the same and communicate them to the University.

12. Learning outcomes

- a. Program Outcomes (PO) - Statements defining the skills, knowledge, and attitude that graduates of a program will be able to demonstrate upon completing the program.
- b. Course Outcomes (CO) - Statements defining the skills, knowledge, and attitude that students will be able to demonstrate upon completing the course. COs are mapped to the POs such that attaining the course outcomes leads to the attainment of program outcomes.
- c. Attainment of COs are mapped to POs such that attaining the course outcomes leads to the attainment of program outcomes

13. Evaluation

For all courses, the evaluation will be based on both formative assessment (Continuous Internal Evaluation, CIE) and summative assessment (Semester End Examination, SEE). Weightage for CIE and SEE will be 60% and 40% respectively.

13.1 Continuous Internal Evaluation (CIE)

Refers to the periodic and continuous *formative assessment* of students' performance during the semester by the teacher(s) of the course to provide timely feedback to students and for guiding "course corrections" by the teachers. The assessment methods may include tests, quizzes, assignments, project evaluations, portfolio evaluations, seminar assessments, etc. CIE will have a weightage of 60% in the determination of the final grading of the course.

13.2 Semester End Examination (SEE)

Refers to a *summative assessment* that covers the entire course syllabus, conducted by the University, at the end of the semester. Appropriate assessment methods aligned with the learning domain and teaching-learning methods are to be used. SEE will have a weightage of 40% in the determination of the final grading of the course.

14. Grading

Course Grade refers to a qualitative measure of performance of a student in each course, based on the percentage of marks secured in Continuous Internal Evaluation (CIE) and Semester End Evaluation (SEE). A Letter grade is awarded for each course.

15. Semester Grade Point Average (SGPA)

Refers to the measure of a student's academic performance in a semester. It is calculated based on the credits and the grades obtained in the courses offered in the semester.

16. Cumulative Grade Point Average (CGPA)

Refers to the measure of the cumulative performance of a student in all the previous semesters and is computed from the 2nd semester onwards. It is calculated based on the credits and the grades obtained in all the courses taken.

17. Academic Bank of Credits (ABC)

The Academic Bank of Credits is a national-level facility for “credit transfer”. It is provided by the Ministry of Education, Govt. of India, to promote the flexibility of the curriculum framework and interdisciplinary/multidisciplinary academic mobility of students across the Higher Education Institutions in the country. The banking and redemption of credits through ABC will be governed by the University's guidelines.

APPENDIX-B

Evaluation Guidelines

CIE and SEE details for various types of courses

1. Theory: PCC/IPCC/PEC/OEC

1.1. Scheme of examinations: CIE + SEE = 60 + 40 = 100 marks

1.2. Continuous internal evaluation (CIE):

1.2.1. CIE (PCC/PEC/OEC)

Type of Questions	Questions to be set (Can have sub- questions a and b)	Questions to Be answered	Marks per question	Total marks
Mid Sem Exam-1				
40-50 % of the total syllabus				
Descriptive Part-1	2	1	10	10
Descriptive Part-2	2	1	10	10
Mid Sem Exam-2				
40-50% of the total syllabus				
Descriptive Part-1	2	1	10	10
Descriptive Part-2	2	1	10	10
Participative Learning: Learning Activity/TASK				
Learning Activities	The Learning Activities/Task comprises class tests/quizzes/assignments conducted for all 5 units/modules each for a max mark of 20. The final task mark is based on the average of all 5 assessments, All tests/quizzes/Assignments are compulsory.			20
Maximum Marks				60

1.2.2 CIE (IPCC/PBL)

Type of Questions	Questions to be set (Can have sub-questions a and b)	Questions to be answered	Marks per question	Total marks
Mid Sem Exam-1				
40-50 % of the total syllabus				
Descriptive Part-1	2	1	10	10
Descriptive Part-2	2	1	10	10
Mid Sem Exam-2				
40-50 % of the total syllabus				
Descriptive Part-1	2	1	10	10
Descriptive Part-2	2	1	10	10
Learning Activities	The learning activity comprises class tests/quizzes/assignments conducted for all 5 units/modules each for a max mark of 20. The final task mark is based on the average of all 5 assessments, All tests/quizzes/Assignments are compulsory.			20
Maximum Marks (A)				60
Minimum Eligibility 50% of Maximum mark (A)=30 marks				
Practical/Project Based Learning (PBL)				
Practical/PBL	Practical/PBL (comprises of implementation of theoretical concepts through projects/problem solving)			60
Maximum Marks (B)				60
Minimum Eligibility 50% of Maximum mark (B) =30 marks				
Maximum CIE Marks [60% of Max marks (A) (Theory)+ 40% of Max marks (B) (Practical/PBL)]				60

1.2.3 Semester End Examination (SEE): 3 Hours Duration

Type of Questions	Module & Teaching hours	Questions to be set	Questions to be answered	Marks per question	Total marks
MCQ	Entire Syllabus	20 (4 questions from each unit/module)	All Questions	1	20
Type of Questions	Module & Teaching hours	Questions to be set (Can have sub-questions a, b, and c)	Questions to be answered	Marks per question	Total eligible max marks
Descriptive	- Unit-1 7-8 teaching hours	2	1	16	16
Descriptive	- Unit-2 7-8 teaching hours	2	1	16	16
Descriptive	- Unit-3 7-8 teaching hours	2	1	16	16
Descriptive	- Unit-4 7-8 teaching hours	2	1	16	16
Descriptive	- Unit-5 7-8 teaching hours	2	1	16	16
				Maximum Marks	100
Minimum SEE Marks to be scored out of 100 marks					40
Minimum SEE Marks to be scored out of 40 marks					16

1.2.4 CIE & SEE for various types of courses

Sl. No.	Courses		Evaluation Scheme						
			CIE (Minimum eligibility marks 50% of Max CIE marks to appear for SEE)		SEE (Minimum eligibility marks 40 % of Max marks)		Total Marks= (CIE+ SEE)	Requirement for securing a final Pass (Letter Grade P)	
			Max Marks	Min eligibility marks 50% of max marks	Max Marks 100 or 50	Minimum eligibility marks required			
40% of max marks	Out of 40 marks	50 % marks (Aggregate of CIE +SEE)							
1	Integrated Professional Core Course (IPCC)		Theory	60	30	100	40	16	100
		Practical	60	30					
		Total (60% of theory +40% of Lab)	60	30					
2	PCC with PBL component	Theory	60	30	100	40	16	100	50
		PBL component	60	30					
		Total (60% of theory +40% of PBL)	60	30					
3	PCC/PEC/OEC		60	30	100	40	16	100	50
4	Laboratory		60	30	50	20	16	100	50
5	Community Engagement		100	50	-	-		100	50
6	Internship Industry/Govt./ MSME/Research center =Total of 4 weeks)		60	30	50	20	16	100	50
7	Capstone Project Research Internship/ Advanced Industry Internship/Project work		120	60	80	32	-	200	100
8	Professional Practice		180	90	120	48	-	300	150

All university examinations (SEE) shall be conducted for a maximum of 100 marks. For assigning the letter grade the university examination marks secured by a student, shall be reduced to 40 marks (except

for serial no.4,5, 6,7 and 8) and added to CIE marks. The minimum marks for securing a passing grade 'C' shall be 50% of the aggregate marks (CIE+ SEE). If the total marks result in a fraction during reduction, it shall be rounded off to the nearest higher value.

2 Laboratory/Practical Course

2.1 Split-up of Marks for evaluation of Practical for 60 CIE marks and 40 SEE marks.

2.2 Split-up of Marks for evaluation of Laboratory work:

2.2.1 Laboratory in-charge faculty will follow rubrics given in the Tables below for an evaluation of laboratory courses

2.2.2 In the case of Practical, the CIE marks shall be based on laboratory observation, records, viva, and at least one practical test.

2.2.3 Continuous Evaluation in every lab session will be done using the format mentioned in the Table to evaluate respective PO's

2.2.4 Rubrics used for Continuous Internal Evaluation of **Laboratory Courses involving experiments with hardware**

Lab Conduction and Record			Lab MSE		
Split-up: 50% (30 Marks) of Maximum CIE marks (60). Each experiment is to be evaluated for conduction with an observation book and record write-up (30 marks per experiment). The final marks for conduction and record are the average of all the specified experiments in the syllabus.			Split-up: 50% (30 Marks) of Maximum CIE marks (60). One test of 30 Marks		
Rubrics per experiment	Marks Distribution	Remarks	Rubrics	Marks Distribution	Remarks
Circuit	02	Evaluation of Record write-up to include weightage for submission on time, neatness, etc.	Write-up	05	
Design	02		Conduction & Viva	15	
Procedure	02			10	
Conduction	06				
Viva	06				
Record write-up	12		Results		
Total Marks	30	Total Marks	30		

2.2.5 Split-up of Marks used for continuous Evaluation of laboratory involving experiments with software (Sample for PCC)

Rubrics for Split up of Marks	Methodology / Process Steps per Experiment	Marks
#R1	Observation, Write up of Procedure / Algorithm/ Program execution, and Conduction of experiment	15
#R2	Viva – Voce	05
#R3	Record writing	10
	Total Marks for each experiment	30
#R4	Internal Test: Lab Internal Assessment	
	(i) Write-up of Procedure/Program/Algorithm	05
	(ii) Conduction/Execution	20
	(iii) Viva-Voce	05
	Total Marks	30

3. Internship and Evaluation

3.1 Introduction

The rise in global competition has prompted organizations to devise strategies to have a talented and innovative workforce to gain a competitive edge. Developing an internship policy is an impactful strategy for creating a future talent pool for the industry. The internship (a form of experiential learning) program not only helps fresh pass-outs in gaining professional know-how but also benefits corporate sectors. The internship also enhances the employability skills of the student passing out from Technical Institutions.

The following list provides a brief illustrative overview of the knowledge, skills, work habits, and character traits commonly associated with 21st-century skills and to be acquired by graduates:

- Critical thinking, problem solving, reasoning, analysis, interpretation, and synthesizing information.
- Scientific literacy and reasoning, the scientific method.
- Research skills and practices, interrogative questioning.
- Creativity, artistry, curiosity, imagination, innovation, and personal expression.
- Information and communication technology (ICT) literacy, media and internet literacy, data interpretation and analysis, and computer programing.
- Oral and written communication, public speaking and presenting, listening.
- Economic and financial literacy, entrepreneurial skills.
- Global awareness, multicultural literacy, humanitarianism.
- Environmental and conservation literacy, ecosystems understanding.
- Civic, ethical, and social-justice literacy.

- Leadership, teamwork, collaboration, cooperation, and facility in using virtual workspaces.
- Perseverance, self-direction, planning, self-discipline, adaptability, initiative.
- Health and wellness literacy, including nutrition, diet, exercise, and public health and safety.

The internship experience will augment the outcome-based learning process and inculcate various attributes mentioned above in a student in line with the graduate attributes defined by the NBA as well as NEP 2020

Following are the intended objectives of internship training.

- (i) Expose Technical students to the industrial environment, which cannot be simulated in the classroom, and hence create competent professionals in the industry.
- (ii) Provide possible opportunities to learn, understand and sharpen the real-time technical/managerial skills required at the job.
- (iii) Expose to the current technological developments relevant to the subject area of training.
- (iv) Use the experience gained from the industrial internship in discussions held in the classrooms.
- (v) Create conditions conducive to the quest for knowledge and its applicability on the job.
- (vi) Learn to apply technical knowledge in real industrial situations.
- (vii) Gain experience in writing reports on technical works/projects.
- (viii) Expose students to the engineer's responsibilities and ethics.
- (ix) Familiarize with various materials, processes, products, and their applications along with relevant aspects of quality control and safety measures.
- (x) Promote academic, career, and/or personal development.
- (xi) Expose the students to future employers.
- (xii) Make students available to the industry for employment.
- (xiii) Understand the psychology of the workers and their habits, attitudes, and approach to problem-solving.
- (xiv) Understand the social, economic, and administrative considerations that influence the working environment of industrial organizations.

3.2 Academic credit framework for the internship and project work undergone as part of the B.Tech. program.

- A minimum of 16 credits to be earned through community engagement activities, Internship and Project work/ professional practice may be counted towards B. Tech. degree program.
- Here, 1 credit is equivalent to a minimum of 40-45 hours of work. Therefore, a full-time intern is expected to spend 40 - 45 hours per week on Internship/Training/Project work etc. This will result in about 640 to 720 hours of total internship and project duration for the B. Tech program.
- To derive the benefits of an internship, it may be introduced in two stages of the B.Tech. program.
- Internships may be full-time or part-time; they are full-time during the summer vacation and part-time during the academic session. The curriculum is flexible to adjust internship

duration. Therefore, opportunities must be provided for experiences that cannot be anticipated when planning the course.

- The departments have the flexibility to schedule internships, Project work, Seminars, etc. according to the availability of the opportunities. However, the suggested minimum requirement regarding Internship duration and credits are as given in Table -B1.

Table-B1. Suggested Credit Framework for Internship and Capstone Project.

Sl. No.	Title	Schedule	Duration	Activities	Credits
1	Community Engagement	Ongoing First-year academic session/ Vacation of 2 nd Semester/ vacation during 3 rd semester (for lateral entry students)	02 weeks	1) Social connect & Responsibility (15 hours lecture to be engaged during second semester) 2) Activity based Internship to focus on the activities pertaining to Social Connect & Responsibility (1 week duration (40 - 45 hours) to be completed during the Vacation of 2 nd Semester/ vacation during 3 rd semester (for lateral entry students) 3) Award of credits in 8 th semester	02
2	Internship	a) Summer vacation after 4 th Semester	02-04 weeks	Industrial/Govt./ NGO/ MSME/ Rural Internship/ Innovation / Entrepreneurship	---
		b) Summer vacation after 6 th Semester	02-04 weeks	Industrial/Govt./ NGO/ MSME/ Rural Internship/ Innovation / Entrepreneurship	
		c) Total of a) and b) should be completed before the beginning of the 8 th semester	04 weeks	Evaluation in 8 th Semester	04
3	Capstone Project work / Professional Practice	7 th Semester	3hours/week	Capstone project phase 1	00
		8 th Semester	16 weeks	Professional Practice /Research Internship/ Capstone project phase 2	10
				Report preparation and writing	
Total Credits					16

Table-B1 states that during the ongoing/ summer vacations after the 2nd Semester, students are required to be involved in Inter/ Intra Institutional community engagement activities as outlined in section 3.3

3.3 Community Engagement

While intra-activities are within the institution, inter-activities shall be between the concerned institution and neighboring institutions. Intra and Inter activities are the activities that are the

impetus to learning techniques. It adds to the comprehensive growth of the mind and associated activities.

As the students are on the verge of learning technical aspects and have a limited period of engagement, it is preferable to expose students to polygonal activities instead of one type of activity. Therefore, activities completed by the students shall not be one type of activity but can be few activities. In this regard, Intra and Inter-Institutional activities shall be completed under the supervision of a faculty on a self-learning basis.

The faculty have to kindle the latent abilities of the students, encourage, guide, supervise and shape them to achieve the desired result. Therefore, a learning agenda in the form of specific learning objectives and outcomes shall be prepared before the start of community engagement. Whatever the activity/activities that are/are done under Intra and Inter-Institutional activities, should ignite the inquisitiveness to learn, enhance the knowledge, thinking ability and imagination, planning, application of mind, execution ability, innovation attitude, listening and understanding, vocabulary, personal expression, public speaking, written communication, oral presentation of the subject matter, acquire leadership qualities and teamwork requirements, responsiveness, ethics, etc.

3.3.1 Societal (Social) related activities

Short-term internships (about 2 weeks) in villages, slums, or urban areas can be under social internship. The internship will be more fruitful if students work in teams. The teams can select one or more fields to do their best in the field of agriculture, watershed management, wastelands development, non-conventional energy, low-cost housing, sanitation, nutrition and personal hygiene, schemes for skill development, income generation, blood bank, and government schemes such as

- i) (Swachh Bharat: Swachh Bharat Mission, Swachh Bharat Abhiyan, or Clean India Mission is a country-wide campaign to eliminate open defecation and improve solid waste management.
- ii) Accessible India: Accessible India Campaign or Sugamya Bharat Abhiyan is a program to serve the differently able community of the country.
- iii) Digital India: A campaign to ensure the Government's services are made available to citizens electronically by improved online infrastructure and by increasing Internet connectivity or making the country digitally empowered in the field of technology.
- iv) Beti Bachao and Beti Padhao: A campaign of the Government of India that aims to generate awareness and improve the efficiency of welfare services intended for girls in India.
- v) Environment and Energy Conservation and Education, legal aid, consumer protection, and allied field including Indian Red Cross Society, National Cadet Corps, Bharat Scouts, and Guides.

Societal activities are one of the NBA graduates attributes that are part of PO6 and PO7, which are reproduced below.

- vi) **PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
- vii) **PO7: Ethics:** Apply ethical principles and commit to professional ethics, human

values, diversity and inclusion; adhere to national & international laws.
Given the above, internship coordinators should encourage students to take up a societal internship as far as possible.

3.3.2 List of proposed activities

- a. Activities concerned with the works of Indian scholars like Charaka and Susruta, Aryabhata, Bhaskaracharya, Chanakya, Madhava, Patanjali, Panini, and Thiruvalluvar, among numerous others
- b. Community Photography.
- c. Short film production: Contemporary aspects, technical aspects, etc.
- d. Internship in Disaster Management.
- e. Solar energy connected activities that help the common man.
- f. Working with Smart City Administration.
- g. Hackathon (it is a design sprint-like event in which computer programs and others involved in software development, including graphic designers, interface designers, project managers, and others, often including domain experts collaborate intensively on software projects).
- h. Industrial Safety, Fire Safety, Electrical Safety, Chemical Process Safety, Food Safety, etc.
- i. Internship and project work in Indian Knowledge System Related Areas/Topics.

3.3.3 Documents to be submitted by Students for Community Engagement Evaluation

3.3.3.1 Student's Diary

The main purpose of writing a daily diary is to cultivate the habit of documenting and to encourage the students to search for details. It develops the student's thought process and reasoning abilities. The students shall record in the daily training diary the day-to-day account of the observations, impressions, information gathered, suggestions given, if any, and activities carried out. It should contain sketches and drawings related to the observations made by the students. The daily training diary should be signed after every day or at least twice a week by the faculty/ in charge of the section (external expert) where the student has been working.

Student's Diary should be submitted by the students along with attendance records. It shall be evaluated based on the following criteria:

- i) Regularity in the maintenance of the diary.
- ii) Adequacy and quality of information recorded.
- iii) Drawings, sketches, and data were recorded.
- iv) Thought processes and recording techniques were used.
- v) Organization of the information

3.3.3.2 Report: Community Engagement (CEN)

After completion of the CEN, the student shall prepare, with a daily diary as a reference, a comprehensive report in consultation with the mentor/s to indicate what he/she has observed and learned in the training period along with the internship outcomes. The CEN report should be signed by the

mentor. The Internship report shall be evaluated based on the following criteria and/or other relevant criteria about the activity completed.

- i) Originality.
- ii) Adequacy and purposeful write-up.
- iii) Organization, format, drawings, sketches, style, language, etc.
- iv) Practical applications, relationships with basic theory, and concepts taught in the appropriate course.
- v) Variety and relevance of learning experience.

3.3.3.3 Procedure for the Evaluation of Community Engagement

- a) Students should submit the reports immediately on completion of the Internship to the respective mentors as outlined in **Table B-2**
- b) The Examination of the Community Engagement will be carried out by the mentor
- c) The Community Engagement shall be slated for 100 marks CIE only and will not have SEE.
- d) Community Engagement marks are based on CIE marks (25 marks for the Social Connect and responsibility, 25 marks for the presentation, and 50 marks for the report and final presentation).
- e) A Viva-Voce examination to be conducted (Presentation followed by question-answer session) and the prescribed credit shall be included in the VIII semester grade card.

3.3.4 Assessment Rubrics for evaluation of Community engagement (Intra and Inter-Institutional Activities)

Table – B2. Community Engagement Assessment Rubrics Scheduled during the first year (Prescribed Period 02 weeks and Prescribed credits: 02)					
Sl No	Sub Activity Head	Performance/ Appraisal	Assessment Rubrics (Allotted Percentage marks decide the letter grade) Percentage Marks	Proposed Document as Evidence	Evaluated by
1	Community Engagement: Inter/ Intra	Excellent	80 to 100	(i) Student's Diary and	Institute Faculty (mentor)
		Good	60 to 79		
		Satisfactory	50 to 59		

	Institutional activities.	Unsatisfactory and fail	< 50	(ii) Community Engagement Report along with the certificate issued from the relevant authorized Authority	together with External Expert, if any.
Note: The total CIE marks are 100 (25 marks for the Social Connect and responsibility, 25 marks for the presentation, and 50 marks for the report and final presentation) shall be the sum of marks allotted to completed activities by the student.					

3.4 Industry Internship (04 weeks) [Scheduled during the intervening period of IV & V semesters and VI & VII semesters]

3.4.1 Industrial Internships

The gap between the theoretical knowledge obtained in the classrooms and the practical skills required in the actual workplace scenarios is fast growing. This has put forth varied challenges to graduating students when it comes to job placements. As institutes cannot have a relevant facility to expose students to a real-time industrial environment, an industrial internship is an appropriate solution.

The main objective of the industry internship is to ensure that the intern is exposed to a real job world environment and gains practical experience. Often, it may be a practical exposure to the theory that has been learned during the academic period. The industry internship helps students understand analytical concepts and tools, hone their skills in real-life situations, and build confidence in applying the skills learned.

3.4.1.1 Industry Internship Benefits

- i) Have ample opportunities to attend seminars, symposiums, workshops, etc. This in turn provides an opportunity to establish rapport with professionals and pioneers in their respective fields for further growth.
- ii) Have wide scope to publish paper/s in journals.
- iii) Good recommendation letter/s that increase the prospectus for further internships, higher studies, and placements.
- iv) Helps to acquire team spirit, motivated acts, techniques to resolve conflicts, etc.
- v) Helps to develop a lot of leadership skills.
- vi) Increases the prospect of placement in the same concern, provided the intern has exhibited a clear understanding of basics and completed the internship.
- vii) Fosters to substantiate the issues with facts and figures.

For AICTE Internship opportunities refer to <https://internship.aicte-india.org/>

3.4.2 Industry Internship Supervision

During the intervening period of the IV & V semesters and VI & VII semesters, students shall be ready for industrial experience. Therefore, they shall choose to undergo an Internship involving Innovation / Entrepreneurship/short-term (about 4

weeks) societal-related activities. Students may choose to work on innovation or entrepreneurial activities, or both resulting in start-up or undergo internship with industry/NGO/ Government organizations/ Micro/ Small/ Medium enterprises to make themselves ready for the industry.

- The internship shall be carried out under the supervision of a faculty mentor. The faculty mentor/guide should,
- Serve as a teacher, mentor, trainer, critic, leader, and boss.
- Provide sufficient time to guide the interns. (Interns are students or a trainee who does a job to gain work experience)
- Play a vital role, along with the Training and Placement Officer, in providing internship opportunities for the students.
- Exhibit qualities such as leadership, strong communication skills, and patience.
- Provide a letter of recommendation in due consultation with students and the industrial organization (if possible) where the internship is intended to be carried out, endorsed by the authority (Principal/Institution Internship Coordinator).
- Each faculty mentor shall supervise the students/Student batches allotted to them. Often, the supervision may be by an external expert also. In such cases, the faculty mentor shall jointly guide the student/s without causing miscommunications/embarrassment to either side.
- Depending on the activity taken up by the students, the internship shall be carried out individually or in batches having not more than three students.
- Faculty Mentor, along with the external expert, shall scrupulously evaluate the work of an individual student or students of a batch and maintain the relevant documents.
- For allotment of CIE marks, the institutions shall prepare the rubrics for each activity offered by the institution as given in Table - B2. The marks shall be allotted by the Internship committee designated by HOD in consultation with the mentors.
- For all activities conducted by the institution, the attendance of the students shall be maintained by the faculty and maintained in their respective departments.

3.4.3 Innovation

Innovation refers to a new or improved product or process or a combination thereof that differs marginally or significantly from the unit's previous product. An innovation center is a place where students are encouraged to implement the innovative ideas formed through imagination, brainstorming sessions, design thinking, and associated activities to bring them to reality. It is a place, where creative minds are shaped.

3.4.4 Entrepreneurship

Entrepreneurship refers to setting up a new business or business and taking on financial risks in the hope of profit. It involves investment to undertake production along with arranging inputs like land, labour, material, and capital, introducing new techniques and products, identifying new sources for the enterprise, etc.

3.4.5 Incubation Center

An organized unit designed for innovation as well as to accelerate the growth and success of new entrepreneurial companies through mentorship and an array of business support resources and services that could include physical space, capital, coaching, common services, and networking connections.

3.4.6 Startup

An entity that develops a business model based on either product innovation or service innovation and makes it scalable, replicable, and self-reliant.

An entity shall be considered a Startup

- i) Up to ten years from the date of incorporation/ registration, if it is incorporated as a private limited company (as defined in the Companies Act, 2013) or registered as a partnership firm (registered under section 59 of the Partnership Act, 1932) or a limited liability partnership (under the Limited Liability Partnership Act, 2008) in India.
- ii) Turnover of the entity for any of the financial years since incorporation/ registration has not exceeded one hundred crore rupees.
- iii) The entity is working towards innovation, development, or improvement of products or processes, or services, or if it is a scalable business model with a high potential for employment generation or wealth creation.
- iv) Provided that an entity formed by splitting up or reconstruction of an existing business shall not be considered a Startup.

3.4.7 Places for Innovation/Entrepreneurial Activities

Students shall carry out Innovation or Entrepreneurial activities or both at the Incubation Center and Entrepreneurship Cell of the parent institution or elsewhere such as ATAL Incubation Centers [A flagship of Atal Innovation Mission (AIM), NITI Aayog for promoting the culture of innovation and entrepreneurship in India], institutes of national importance, public sector units, IT companies, government organizations, and non-governmental organizations, industries including MSME, etc.

- **Institutes should deter students to opt for internships at places established for commercial benefits.**

3.4.8 Assessment Rubrics for Innovation / entrepreneurship/ Industry Internship Activities (refer Table B3)

Once the internship begins, the students are required to maintain a diary/journal and submit a report regularly to the guide. These reports should summarize the activities in which the student was involved during the previous week's period. At the end of the internship, each student is required to submit a hard copy of the consolidated diary/journal and report for evaluation. The report should indicate the learning and

achievements of the internship.

Table – B3. Innovation/entrepreneurship/ Industry Internship Activities and Assessment Rubrics				
Scheduled during the intervening period of IV & V semester and VI & VII Sem (Prescribed Period 04 weeks: Credits 04)				
Sub Activity Head	Performance/ Appraisal	Assessment Rubrics	Proposed Document as Evidence	Evaluated by
(1) Development of new product/ Business Plan/ registration of start-up	Excellent	80 to 100	(i) Student’s Diary and (ii) Internship Report or the activity report along with Certificate or Declaration from relevant Authorized Authority. Wherever only Certificate is issued, Assessment shall be at the institute as per (i) and (ii) to decide the letter grade.	(i)Institute Faculty (mentor) together with External Expert if any.
	Good	60 to 79		
	Satisfactory	50 to 59		
	Unsatisfactory and fail	< 50		
(2) Internship with Industry/ Govt. / NGO/ PSU/ Any Micro/ Small/Medium Enterprise.	Excellent	80 to 100		
	Good	60 to 79		
	Satisfactory	50 to 59		
	Unsatisfactory and fail	< 50		
Note: (i) The total CIE marks shall be the sum of marks allotted to successfully completed activities by the student.				

4. Capstone Project/Professional Practice / Research Internships

4.1 Capstone project/Research Internship/Professional practice of sufficient duration encourages students early on in their careers. Its main goal is to allow improving their analytical and technical skills in an international environment. An internship can be in an industry or at an appropriate workplace.

- Research internships/industrial projects have different purposes and come with a set of benefits. A prior experience in any field is always preferred over a fresh start. Therefore, one of them can be selected depending on the interest the students have. Internships pose unexpected challenges and make students think appropriately, tackle difficulties with ease, and act in a scholarly way to get past the hurdles and practical constraints. An internship is always beneficial however good or bad it is.
- Professional practice not only enhance one's learning but also identifies him/her as someone who commits to approaching a project and completing it with or without guidance. Internship learning is an impetus for professional development.
- While a research internship is a stepping stone to higher studies, an industry internship is a pathway to a placement. Those who are self-motivated and interested in searching for new things that are original and unique can choose a research internship. Those who are interested in real industry- experience and aspire to get a job soon after graduation can choose an industry internship.
- Research Internships (Also known as dissertation internships) are focused research projects that push students' intellectual abilities beyond those driven by the classroom. Often, a research internship typically helps solve problems that are usually part of major

research projects. It involves a short theoretical or experimental research project supervised by a researcher.

- The research internships, under the advice of a faculty supervisor, can be one's own selected project or a project on which a Researcher is researching, or a new project/real-world project offered by an organization. The research area may be about single or multidisciplinary fields such as science, technology, engineering, mathematics, management, and business studies. Research internships can be carried out either individually or in teams (not exceeding 3 or 4 students).
- Research internship opportunities, before graduation, maybe in a laboratory of college, a research institute, or a company's R & D department. Apart from fixed working hours of the day of an organization, the researcher can devote sufficient time to other research-related activities for early and successful completion of the Research Internship.

4.2 Necessary Skills for Research Internship and Industrial Internship

For the internships to progress without hurdles and for successful completion, the Researchers should maintain a harmonious relationship with the guide/s, administrators, co-workers, and others, and strictly adhere to the rules and regulations of the workplace.

The other skills required or acquirable during the Internship are,

- Good Communication skills.
- Attention to detail.
- Planning and scheduling.
- Documentation.
- Critical thinking.
- Data collection.
- Data analysis.
- Ability to maintain quality, safety, and/or infection control standards.
- Appreciating and practicing ethical issues.

4.3 Responsibilities of an Intern

Interns,

- a) If working with a researcher, shall assist the researcher in an ongoing research project or work collaboratively in designing a new project of mutual interest.
- b) Shall engage in literature survey and get an insight of the research work at the initial stages.
- c) Shall compile data, sort, file, implement ideas with minimal guidance and assist write papers.
- d) Shall become familiar with several tools [meters (Electrical and Electronics, mechanical, computer, etc.)) used in data collection, software, graphic software, Statistical Package for the Social Sciences (SPSS) software [IBM's statistical software platform], etc.
- e) Shall attain skills with Microsoft Word Office, Excel, PowerPoint, Outlook, etc.
- f) Shall give a mid-term oral presentation to a committee for review and feedback.
- g) Shall attend discussions, meetings, symposiums, classroom lectures, etc., to learn new scientific techniques, design experiments, analyze results, and formulate different hypotheses.
- h) Shall learn to write reports and be able to correspond independently.
- i) Shall manage time effectively.

- j) Shall keep a track of the progress of the project.
- k) Shall develop integrative thinking.

4.4 Research internship Outcomes

- a) Generating technical paper/s and publishing in refereed journal/s.
- b) Possibility of acquiring intellectual ownership and patent.
- c) Build a prototype for an idea on which the research was carried out.
- d) File patent/s.
- e) Add academic knowledge to the field.
- f) Enhanced ability in arranging meetings, presentations, seminars, training, etc.
- g) Improved conscientiousness and ethics.

4.5 Research internships Benefits

- a) Are a great way to pursue an academic career in teaching and research, as a Research Scientist at a Research Organization, Company, Industry sector, etc.
- b) Establish professional networks for a future career.
- c) Pave the way to join a research team and work alongside leading experts in the field.
- d) Introduced to new ideas through interaction with like-minded students and others.
- e) Develop research skills and knowledge in a specific area of interest.
- f) Provide opportunities for growth, achievement, and personal development.
- g) Offer an opportunity to publish a research paper that will boost the resume while applying for Post Graduate Studies

5. Evaluation Procedure of Rresearch Internship /professional practice /and Capstone Project (16 weeks)

- 5.1** The students pursuing the course Research **Internship /professional practice /and Capstone Project** shall submit the diary recordings of day-to-day activities to the concerned guide, reporting progress achieved in the course and seeking guidance to proceed with the internship. The interns should provide all the details to the guide so that he/she can discuss with the employer to make the internship successful.
- 5.2** The intern should constantly update the guide about the progress of the internship. The guide should know the intern's internship tasks, duties, responsibilities, and potential projects. The evaluation of interns and their internship progress should be honest and constructive.
- 5.3** The hardcopy or softcopy of the diary maintained by the interns must be signed at regular intervals by the guide.
- 5.4** Regarding the intern's feedback, the guides should propose changes in internship activities so that they are helpful to the internship.
- 5.5** Illustrations, drawings, photos, forms, samples, classified materials, etc., are to be included in the report only after obtaining the consent of the concerned authorities and should indicate the source of all such material. The final report should also be submitted to the place where the internship was carried out. The report should avoid a tone that is predominantly cynical or unduly critical of the employer or of those with whom the student intern has worked. The content of the report must be based on interns' own work.

5.6 Recommendation letter

The guide must state whether the intern,

- a) Exceeded the expectations of the internship.
- b) Met the expectations of the internship.
- c) Did not meet the expectations of the internship.
- d) Did work to a satisfactory level.
- e) Did an unsatisfactory internship.

In the end, the guide should issue a recommendation letter.

5.7 Continuous Internal Evaluation (CIE)

The guides should evaluate the interns using the following as well as any other appropriate methods.

- a) Punctuality of intern.
- b) Conduct and character.
- c) Tactfulness and politeness with colleagues and the public.
- d) Attitude regarding professionalism.
- e) Inquisitiveness and eagerness to learn.
- f) Research attitude.
- g) Problem-solving techniques.
- h) Innovation mindset.
- i) Time management and meeting deadlines.
- j) Receptiveness to feedback and critiques.
- k) Ability to work in a team as a member.
- l) Ability to work without supervision.
- m) Supervisory skills and leadership skills.
- n) Judgment and decision-making skills.
- o) Writing skills, oral communication skills, technical communication skills, computer skills, analysis skills, and business writing skills.
- p) Appropriateness of technical skills.
- q) Familiarization with writing technical papers, standards, codes, etc.
- r) Reading Behavioural attitude.
- s) Outcomes.
- t) Successes and failures experienced

5.8 Assessment of CIE marks

- 5.8.1 **Single discipline:** The CIE marks shall be awarded by a committee consisting of the Head of the concerned Department and two faculty members of the Department, one of whom shall be the Guide. The CIE marks awarded for the internship, shall be based on the evaluation of the diary, presentation skill, and viva-voce in the ratio of 50:25:25.
- 5.8.2 **Interdisciplinary:** The CIE marks awarded for the internship, shall be group-wise at the institution level with the participation of all guides of the internship. Participation of external guide/s, if any, is desirable.
- 5.8.3 The CIE marks awarded for the internship, shall be based on the evaluation of the diary, presentation skill, and viva-voce in the ratio of 50:25:25.

5.9 Assessment of SEE marks

- 5.9.1 **Single discipline:** Contribution to the internship and the performance of each group member shall be assessed individually in the semester-end examination (SEE) conducted at the department. Marks shall be awarded based on the evaluation of the report, presentation skill, and viva-voce in the ratio of 50:25:25.
- 5.9.2 **Interdisciplinary:** Contribution to the internship and the performance of each group member shall be assessed individually in the semester-end examination (SEE) conducted separately at the departments to which the student/s belongs. Marks shall be awarded based on the evaluation of the report, presentation skill, and viva-voce in the ratio of 50:25:25.

5.10 Evaluation Industry Internship/Professional Practice /Project Work:

Split-up of marks for evaluation of Project work for 100 CIE marks and 100 SEE marks.

Split up	Rubrics		% Marks
Report (50 % Marks)	Content Development	Abstract/ Synopsis Write-up	10
		Selection of Topic/ Relevance of the subject to the concerned discipline	05
		Problem Identification	05
		Objectives and Methodology	05
	Problem-Oriented Exposition	Literature Survey (Papers/Sites/Sources Surveyed)	10
		Documentation/ Systematic Approach	10
		Results (with inferences, Conclusions, etc.)	05
Project Presentation Skill		Quality of preparation of presentation	05
		Communication Skills	05

(25 % Marks)		Technical knowledge and awareness	05
		Individual involvement	10
Viva- Voce (25 % Marks)		The clarity in answering questions relating to fundamentals and concepts	10
		The clarity in answering the questions related to the project	05
		The understanding ability of the questions asked	05
		The confidence in answering the questions asked.	05
		Total Marks	100 %

B. Tech. Regulations and Syllabus

Vision:

To achieve excellence in engineering education by creating a dynamic learning environment, fostering collaboration and interdisciplinary research in Artificial Intelligence and Data science, while empowering our graduates to become leaders in the global AI landscape.

Mission:

- Creating an ecosystem of academic excellence by incorporating the best possible teaching-learning methodology, collaborative research and the usage of modern IT infrastructure and tools.
- Grooming professionals with high ethical values and inculcate the ability to solve real-life problems that involves data analytics.
- Contribute towards the innovation of computing, AI system and Data Science to raise satisfaction level of all stakeholders.
- To prepare professionals for employment in industry, research, higher education, community partnerships, and entrepreneurship to benefit the society.

Program Educational Objectives (PEOs):

Program educational objectives are the broad statements that describe the career and professional accomplishments that the program is preparing graduates to achieve.

1. To have a successful professional career with capabilities to build innovative solutions by applying the knowledge of Data Science and using technology as a tool to solve real-world problems.
2. To develop an ethical attitude and shall exhibit effective skills in communication, management, teamwork, and leadership to lead and work as a team in a professional environment.
3. To develop engineering, problem-solving, and critical thinking skills to create social, economical and sustainable impact and develop an ability to adapt themselves to the dynamically changing technologies for catering to the organizational needs.

Program Outcomes (POs):

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, making effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs):

1. Apply the principles of artificial intelligence and data science to develop the systems that require data analysis, inference, perception, knowledge representation, and learning.
2. Obtain the competencies to excel in Employment, Higher studies and Research in Artificial Intelligence and Data Science while upholding ethical values.
3. Demonstrate proficient professional abilities for seamless collaboration within diverse and interdisciplinary teams, embracing a growth-oriented mindset.

Scheme & Syllabus for
First Year B.Tech.(AIDS, AIML, CCE, CSE, ISE, RAI, CYB, AND CSBS)
2026-27

ADVANCED CALCULUS

Course Code:	26MAT103	Course Type:	BSC
Teaching Hours/Week (L-T-P):	3-1-0	Credits:	04
Total Teaching Hours:	45+15+0	CIE + SEE Marks:	60+40

Teaching Department: Mathematics

COMMON FOR CSE/ISE/CCE/AIM/AI&DS/R&AI/CYB/CBS

Course Objectives:

- This course will enable the students to master the concepts, methods of solutions and basic tools used under single variable, multivariable differential calculus.
- The course also imparts knowledge of modeling and solving Ordinary and partial differential equation.
- The course also includes techniques of multiple integration with applications

UNIT-I

Differential calculus	9+3 Hours
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Polar curves, angle between tangent and radius vector, angle of intersection, pedal equation, radius of curvature (no derivations). Taylor's and Maclaurin's series (without proof)-problems.

UNIT-II

Partial Differentiation	9+3 Hours
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Partial derivatives, Homogeneous function, Euler's theorem, Total derivative, Partial differentiation of implicit and composite functions, Jacobian, Maxima and Minima for function of two variables. Lagrangian multipliers,

UNIT-III

Ordinary differential equation	9+3 Hours
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Linear differential equation, Bernoulli differential equation, exact differential equation, second and higher order differential equation with constant coefficients-complimentary function, particular Integral, variation of parameter.

UNIT-IV

Partial differential equations	9+3 Hours
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Formation of partial differential equations by eliminating arbitrary constants and arbitrary functions. Classification of 2nd order PDES. Solution of P.D.E: Direct integration method and method of separation of variables. Solution of heat, wave and Laplace equations.

UNIT-V

Multiple Integrals	9+3 Hours
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Multiple integrals –Evaluation of double and triple integrals, change of variables, area using double and volume using triple integrals. Gamma and Beta functions, Relation between Beta and Gamma functions, properties and Problems

Course Outcomes: At the end of the course student will be able to

1	Apply concepts of polar curves and expand functions into Taylor series.
2	Apply concept of multivariate calculus and Extremize functions arising in engineering field.
3	Model and solve Ordinary differential equations in engineering field
4	Form and solve partial differential equations relevant to engineering.
5	Apply concepts of multiple integration and special functions.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓		
↓ Course Outcomes														
26MAT103.1	3	2	-	-	-	-	-	-	-	-	-	-	-	-
26MAT103.2	2	2	-	-	-	-	-	-	-	-	-	-	-	-
26MAT103.3	3	2	-	-	-	-	-	-	-	-	-	-	-	-
26MAT103.4	3	2	-	-	-	-	-	-	-	-	-	-	-	-
26MAT103.5	3	2	-	-	-	-	-	-	-	-	-	-	-	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1	B.S. Grewal, “Higher Engineering Mathematics”, Khanna Publications, 43 rd Edition, 2015.
2	Erwin Kreyszig, “Advanced Engineering Mathematics”, John Wiley and Sons, 10 th Edition (Reprint), 2016.
3	N.P. Bali and M.Goyal, “A text book of Engineering Mathematics”, Laxmi Publications, 2010.

REFERENCE BOOKS:

1	G.B. Thomas and R.L.Finney, “Calculus and Analytic geometry”, Pearson, 2002.
2	W.E. Boyce and R.C. DiPrima, “Elementary Differential Equations and Boundary Value Problems”, Wiley India, 2009
3	G.F. Simmons and S.G. Krantz, “Differential Equations”, McGraw Hill, 2007
4	Shepley L Ross, “Differential equations” 3 rd edition, Wiley, 2004
5	Shanthi Narayan, “Differential Calculus, 6 th edition, Shyam Lal Charitable Trust, Delhi.

E Books / MOOCs/ NPTEL

1	http://nptel.ac.in/courses/111106100/
2	https://nptel.ac.in/courses/122101003

QUANTUM COMPUTING AND MODERN PHYSICS (For all IT Branches and RAI)			
Course Code:		26PHY102	Course Type: BSC
Teaching Hours/Week (L:T:J:SL):		3:0:2:0	Credits: 04
Total Teaching Hours:		45+0+30+0	CIE + SEE Marks: 60+40
Teaching Department: Physics			
Course Objectives:			
1.	To understand the concepts of quantum mechanics.		
2.	To study the principles of quantum computing and its applications.		
3.	To study the properties of conductors and superconductors and their applications.		
4.	To identify the fundamental principles of semiconductors to apply in electronic devices.		
5.	To understand the principles, construction, and applications of LASERs and Optical fibres.		
UNIT-I			
Quantum Mechanics			09 Hours
Introduction to quantum mechanics, de Broglie Hypothesis and Matter Waves, de Broglie wavelength, Phase Velocity and Group Velocity, Relation between Phase Velocity and Group Velocity (Derivation), Relation between Particle Velocity and Group Velocity, Heisenberg’s Uncertainty Principle - Application of Uncertainty Principle (nonexistence of electron inside the nucleus), Wave Function, Physical Significance of a wave function, Time independent Schrödinger wave equation (Derivation), Eigen functions and Eigen Values, Particle inside one dimensional infinite potential well, Numerical Problems.			
UNIT-II			
Quantum Computing			09 Hours
Introduction to Quantum Computing, Moore’s law and its end, Differences between Classical & and Quantum computing. Linear vector space, linearly dependent and independent vectors, Hilbert space, Concept of qubit: Dirac representation and its properties. Representation of qubit by Bloch sphere. Single and Two qubits. Extension to N qubits, Differences between Classical & and Quantum bits. Matrix operations: Matrix representation of 0 and 1 States, Identity Operator I, Applying I to $ 0\rangle$ and $ 1\rangle$ states, Row and Column Matrices and their multiplication (Inner Product), Orthogonality, Orthonormality. Quantum Gates: Single Qubit Gates: Pauli – X, Y and Z Gates, Hadamard Gate, and Phase Gate (only S Gate). Numerical Problems.			
UNIT-III			
Conductors and Superconductors			09 Hours
Free-electron concept. Classical free-electron theory – Assumptions, Drift velocity, mean collision time, mean free path and relaxation time, Expression for electrical conductivity in metals (Drude-Lorentz theory), Failures of classical free-electron theory. Quantum free - electron theory – Assumptions & success, Fermi energy and Fermi factor for metals and its variation with temperature, Numerical problems. Introduction to superconductors, characteristic properties: Meissner effect, Critical field, Critical Current. Type-I and Type-II superconductors. BCS theory (qualitative). High temperature superconductors, Numerical problems.			

UNIT-IV	
Semiconductors	09 Hours
Introduction to semiconductors, intrinsic and extrinsic semiconductors - carrier generation, Direct and indirect band gap semiconductors. Expression for concentration of electrons in conduction band and holes concentration in valance band (only mention the expression), Fermi level in Intrinsic and Extrinsic Semiconductors and its behavior with temperature, Electrical conductivity of an intrinsic semiconductor (derivation) and effect of temperature on intrinsic conductivity, Hall effect, Expression for Hall coefficient (derivation) and its application. p-n junction: Junction formation, Unbiased and biased p-n junction, Devices: LED, Photodiode and solar cell. Numerical problems.	
UNIT-V	
LASER and Optical Fibers	09 Hours
Introduction to lasers, Characteristics of LASER, Interaction of radiation with matter, Einstein's coefficients, Expression for energy density in terms of Einstein coefficients, Requisites of a Laser System. Conditions for Laser action. Principle, Construction and Working of Nd:YAG LASER and semiconductor LASER, Application of Lasers in Bar code scanner and Laser Printer. Numerical Problems. Introduction and Principle of Optical Fibers (TIR), Propagation mechanism in optical fibers - Angle of Acceptance and Numerical Aperture N.A.), Expression for NA, Fractional Index Change, Number of Modes and V Number, Types of Optical Fibers based on refractive index profile & mode guiding, Attenuation, Factors contributing and Mention of Expression for Attenuation coefficient, Numerical problems.	
List of Experiments	
1.	Energy gap of a semiconductor (any method).
2.	Hall effect
3.	I-V characteristics of Zener diode
4.	Solar cell characteristics.
5.	Determination of wavelength by Laser diffraction.
6.	Diffraction of White light by optical grating
7.	Determination of acceptance angle and numerical aperture of the given Optical Fiber.
8.	Photo electric effect – Determination of the work function of the material of the emitter of a photocell.
9.	Photo-Diode characteristics
10.	LED characteristics and determination of Planck's Constant using LEDs.
11.	Transistor Characteristics.
12.	Determination of Fermi Energy of Copper.
13.	Resonance in LCR circuits.
14.	Simulation experiment.
Course Outcomes: At the end of the course students will be able to	
1.	Apply the knowledge of quantum mechanics to analyse the physical properties exhibited by particles at sub-atomic level.
2.	Apply the basic principles of Quantum Mechanics in Quantum Computing.
3.	Apply knowledge of conductor and superconductor properties to engineering applications.

4.	Analyze the role of semiconductors in electronic devices, exploring the properties of semiconductors.
5.	Analyze the properties of optical waves in the phenomena of lasing action and signal propagation.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓
↓ Course Outcomes												
26PHY102-1.1	3	2						2			1	
26PHY102-1.2	3	2						2			1	
26PHY102-1.3	3	2						2			1	
26PHY102-1.4	3	2						2			1	
26PHY102-1.5	3	2						2			1	

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Parag K Lala, "Quantum Computing – A Beginner's Introduction", Indian Edition, McGraw Hill, Reprint 2020.
2.	B. G. Streetmann, "Solid State Electronic devices", 6 th edition, Prentice Hall India Learning Private Limited.
3.	A. Ghatak, "Optics", Tata McGraw Hill Pub., 5 th Edition, 2012.

REFERENCE BOOKS:

1.	Michael A. Nielsen & Isaac L. Chuang, "Quantum Computation and Quantum Information", Cambridge Universities Press, 2010 Edition.
2.	Vishal Sahani, "Quantum Computing", McGraw Hill Education, 2007 Edition.
3.	Maria Luisa Dalla Chiara, Roberto Giuntini, Roberto Leporini, Giuseppe Sergioli, "Quantum Computation and Logic: How Quantum Computers Have Inspired Logical Investigations", Trends in Logic, Volume 48, Springer.
4.	Gupta and Kumar, "Solid State Physics", K. Nath & Co., Meerut.
5.	A. J. Dekker, "Electrical Engineering Materials", Prentice Hall India Pub., New Delhi, Reprint 2011.
6.	S. O. Pillai, "Solid State Physics", New Age International Private Limited, 8 th Edition, 2018.
7.	M. Ali. Omar, "Elements of Solid State Physics: Principles and Applications", Pearson Publishers.
8.	Arthur Beiser, "Concepts of Modern Physics", Tata McGraw Hill Education Private Limited, Special Indian Edition, 2009.
9.	Kenneth Krane, "Modern Physics", Wiley International, 3 rd Edition, 2012.
10.	Michael Tinkham, "Introduction to Superconductivity", II Edition, McGraw Hill, INC

E Books / MOOCs/ NPTEL/ Web links

1.	LASER: https://www.youtube.com/watch?v=WgzynezPiyC
2.	Superconductivity: https://www.youtube.com/watch?v=MT5Xl5ppn48
3.	Optical Fiber: https://www.youtube.com/watch?v=N_kA8EpCUQo
4.	Quantum Mechanics: https://www.youtube.com/watch?v=p7bzE1E5PMY&t=136s
5.	Quantum Computing: https://www.youtube.com/watch?v=jHoEjvuPoB8
6.	Quantum Computing: https://www.youtube.com/watch?v=ZuvCUU2jD30
7.	Physics of Animation: https://www.youtube.com/watch?v=kj1kaA_8Fu4
8.	Statistical Physics Simulation: https://phet.colorado.edu/sims/html/plinko-probability/latest/plinkoprobability_en.html
9.	NPTEL Superconductivity: https://archive.nptel.ac.in/courses/115/103/115103108/

10.	NPTEL Quantum Computing: https://archive.nptel.ac.in/courses/115/101/115101092
11.	Virtual LAB: https://www.vlab.co.in/participating-institute-amrita-vishwa-vidyapeetham
12.	Virtual LAB: https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1
Activity Based Learning (Suggested Activities in Class)/ Practical Based learning	
1.	http://nptel.ac.in
2.	https://swayam.gov.in
3.	https://virtuallabs.merlot.org/vl_physics.html
4.	https://phet.colorado.edu
5.	https://www.myphysicslab.com

PROBLEM SOLVING THROUGH PROGRAMMING

Course Code:	26CSE103	Course Type:	ESC
Teaching Hours/Week (L:T:P:J:SL)	3:0:2:0:2	Credits:	04
Total Teaching Hours:	45:0:30:0:30	CIE + SEE Marks:	60+40

Teaching Department: Any

Course Objectives:

1	Make students learn the basics of C programming language including the basic data types, Operators and Evaluating expressions in C.
2	Apply the concepts of decision making and looping in problem solving to demonstrate its usage using simple programs.
3	Design and implement efficient C programs using arrays and strings to manage and manipulate data effectively.
4	Demonstrate the usage of Pointers and User-defined functions, code reusability in problem solving along with parameter passing and returning with the help of user defined functions.
5	Demonstrate the usage of structures and File handling that are essential for understanding the concepts with simple examples.

UNIT-I

Introduction To Computer and C language, Operators and Expressions:	09 Hours
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Introduction to computers: Introduction to Computer System and problem solving, Program Development steps, Introduction to Programming Languages, input and output devices, Algorithm, Flowchart, Designing efficient programs. Evolution & Characteristics of C Language, Structure of a C Program, C Compilation Model. Characters set, C tokens, Keywords and identifiers, Constants, Variables, Data Types, and Declaration of Variables. Formatted and Unformatted Input/Output:Formatted I/O: printf() ,scanf() ,Unformatted I/O:getchar(),putchar() ,gets() and puts(), Command line arguments.

Operators and Expressions: Arithmetic operators, Relational operators, Logical operators, Assignment operators, Increment, Decrement operators, Conditional (Ternary) operator, Bitwise operators, Special Operators. Arithmetic expressions, Evaluation of expressions, Precedence of operators, Type conversions in expressions, Operator precedence and associativity.

UNIT-II

Decision control and Looping statements:	09 Hours
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Decision making and Branching: Decision making with if statement, Simple if Statement, the if...else statement, Nesting of if...else statements, The else...if ladder, The switch statement.

Decision making and Looping: The for statement, the while statement, the do...while statement, Jump statements (goto, break, continue).

UNIT-III

Arrays and Strings:	09 Hours
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Arrays: Declaration of arrays, accessing the elements of an array, storing values in arrays, Operations on arrays, Passing arrays to functions, Multidimensional arrays-Two dimensional arrays, operations on two-dimensional arrays, two-dimensional arrays to functions, Example Programs on arrays:linear search, binary search, selection sort, bubble sort.

Strings: Declaring and Initializing strings, String manipulation functions, String input and output functions.

UNIT-IV

Pointers, Functions:	09 Hours
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Introduction to pointers: declaration and initialization of pointers, accessing the address of the variable, accessing the variable through the pointer.

Functions: Introduction using functions, Function definition, function declaration, function call, return statement, passing parameters to functions-call by value, call by reference, storage classes, scope of variables, recursive functions, Example Programs on Functions.

UNIT-V											
Structures, Files:	09 Hours										
Structures: Defining Structure, Declaration of Structure Variable, Accessing Structure members, copying and comparing structure variable, operation on individual member, nesting of structures, Array of structures. Application of pointers and function on Structures. Files: Basic file operations: Open, Close, Read, Write.											
List of Experiments											
Part-A											
1.	Write a C program to find the roots of a quadratic equation $ax^2+bx+c=0$.										
2.	Write a C program to find the sum of all the digits and occurrence of a digit in the number.										
3.	Write a C program to find the GCD and LCM of given two numbers using Euclid's method.										
4.	Write a C program to print the prime numbers in a given range.										
5.	Write a C program to perform a binary search for a given key integer in a single dimensional array of numbers in ascending order and report success or failure in the form of a suitable message.										
6.	The books in the library are randomly placed on the shelves. Design a C program that sorts the books based on ISBN, Use bubble sort to implement the program.										
7.	Write a C program to find if a given string is a palindrome or not using string manipulation functions.										
8.	An event registration system takes inputs from the command line: Participant Name, Age, and Event Name. Write a C program to display the details.										
Part-B											
1.	Design, develop and execute a program in C to read two matrices A(M x N) and B(p x q) and compute the product of A and B.										
2.	Write a C program using functions readmat(), rowsum (), colsum (), totsum () and printmat() to read the values into a two dimensional array A, find the sum of all the elements of a row, sum of all the elements of a column, find the total sum of all the elements of the two dimensional array A and print the results.										
3.	Write a C program to input N integer numbers into a single-dimensional array, sort them into ascending order using selection sort technique, and then to print both the given array and the sorted array with suitable headings.										
4.	Write a C program to read N integers into an array A and find the sum of elements using pointers.										
5.	Write a C program to perform a linear search for a given key integer in a single-dimensional array of numbers and report success or failure in the form of a suitable message using functions.										
6.	Write a C program to enter the information like name, register number, marks in 6 subjects of N students into an array of structures, and find the average & display grade based on average for each student. <table border="1" data-bbox="223 1518 742 1729"> <thead> <tr> <th>Average</th><th>Grade</th></tr> </thead> <tbody> <tr> <td>80-100</td><td>Distinction</td></tr> <tr> <td>60-79</td><td>First Class</td></tr> <tr> <td>40-59</td><td>Second Class</td></tr> <tr> <td><40</td><td>Fail</td></tr> </tbody> </table>	Average	Grade	80-100	Distinction	60-79	First Class	40-59	Second Class	<40	Fail
Average	Grade										
80-100	Distinction										
60-79	First Class										
40-59	Second Class										
<40	Fail										
7.	A college department wants to store and retrieve student names and marks for an internal audit. Write a C program that saves the data to a file named students.txt and then reads and displays the contents to verify the information.										
8.	Design a C program that accepts two numerical values and an arithmetic operator (+, -, *, /) as command-line arguments. The program should compute and display the result of the specified operation.										

Course Outcomes: At the end of the course student will be able to:																																																																																																
1	Describe the basics of computer systems, basics of C and the process of problem-solving aspects using algorithmic solutions for a given problem.																																																																																															
2	Apply the knowledge to develop the C program using control statements such as branching and looping constructs for a given problem.																																																																																															
3	Demonstrate the ability to implement and analyze operations on arrays and strings to solve computational problems.																																																																																															
4	Apply the concept of pointers along with code reusability, parameter passing, and return values to develop efficient and maintainable C programs using functions.																																																																																															
5	Implement and demonstrate the use of structures and file handling mechanisms in C programs.																																																																																															
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2	Byron Gottfried, Programming with C, Schaums outlines series, 4th edition, 2018.																																																																																															
3	Computer fundamentals and programming in C, Reema Thareja, Oxford University, 3rd edition, 2023.																																																																																															
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2	Rajiv Khanna, Computer Concepts and C Programming, New Age International Pvt Ltd Publishers, 2nd Edition, 2011.																																																																																															
3	Kernighan & Ritchie, “The C Programming (ANSI C)”, Prentice Hall; 2nd Edition, 1998.																																																																																															
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1	http://www.lysator.liu.se/c/bwk-tutor.html#introduction																																																																																															
2	C programming Tutorial by Mark Burgers http://markburgess.org/CTutorial/C-Tut-4.02.pdf																																																																																															
3	http://nptel.ac.in/courses/106105085/4																																																																																															

BASIC ELECTRONICS			
Course Code:	26ECE111	Course Type:	ESC
Teaching Hours/Week (L: T:P: S):	3-0-0-0	Credits:	3
Total Teaching Hours:	45 Hrs	CIE + SEE Marks:	60+40
Teaching Department: Electronics and Communication Engineering			
Course Objectives:			
1	Familiarize the student with Semiconductor devices like Diodes with simple applications such as Rectifiers.		
2	Familiarize the student with Semiconductor devices like Transistors with simple applications such as switch and amplifier circuits.		
3	Analyze the working of simple electronic circuits involving Op-amps and Linear Regulator ICs.		
4	Understand the concepts of feedback and analyze Feedback amplifier and Oscillator circuits.		
5	Explore the fundamentals of Modern communication and Embedded Systems.		
UNIT-I			
Diodes and their Applications			09 Hours
Semiconductor Diode, Diode Equivalent circuits, Half Wave Rectifier, Full wave Bridge Rectifier, capacitor filter circuit (only qualitative approach). Zener Diode and its use in Voltage Regulation.			
UNIT-II			
Transistors and their Applications			09 Hours
Bipolar Junction Transistor: (npn only) Construction and operation, Common Emitter Characteristics, CE-RC coupled amplifier (frequency response excluded), BJT as a switch. Field Effect Transistor: Construction Drain and Transfer Characteristics of n-channel Enhancement mode MOSFET, CMOS Inverter.			
UNIT-III			
Op-Amp & Linear IC Applications			09 Hours
Introduction, Op-Amp Specifications, Differential & Common-Mode operation, Op-Amp applications: Inverting/Non- Inverting Amplifier, Inverting Summer, Integrator, Differentiator, Comparator.78XX series IC Voltage Regulators.			
UNIT-IV			
Feedback and Oscillator Circuits			09 Hours
Feedback– Principle and advantages of negative feedback, Voltage series feedback amplifier, Concept of positive feedback, Op-Amp Oscillators – RC phase shift, Hartley and Colpitts’s Oscillator.			
UNIT-V			
Fundamentals of Communication and Embedded Systems			09 Hours
Modern communication system scheme (Block scheme), Information source, Input Transducers, Transmitter, Channels, Receivers, Noise, Fundamentals of Cellular communication. Embedded system definition, Elements of Embedded systems, Core of Embedded systems, Microprocessor v/s Microcontroller, Sensors and Actuators with Examples.			
Course Outcomes: At the end of the course student will be able to:			
1	Design diode-based circuits and voltage regulators		
2	Analyze the operation of transistors and CMOS inverters		
3	Apply operational amplifiers in analog signal processing		
4	Evaluate the effects of feedback and oscillator circuits		
5	Describe the components of modern communication and embedded systems		

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓	
↓ Course Outcomes												1	2
26ECE111.1	3	2	1								1	2	
26ECE111.2	3	2	1					1				2	
26ECE111.3	3	2	1									2	
26ECE111.4	3	2	1		1						1	2	
26ECE111.5	3	2	1									2	

1: Low, 2: Medium, 3: High

TEXTBOOKS:

1	“Electronic Devices and Circuits” David A. Bell, PHI, 5 th , 2014
2	“Electronic Devices and Circuit Theory” Robert L. Boylestad, Louis Nashelsky, Pearson, 11 th Edition, 2016.
3	“Op-Amps and Linear Integrated Circuits”, Ramakanth A Gayakwad, Pearson, 4 th Edition, 2016.
4	“Introduction to Analog and Digital Communications “Simon Haykin, Wiley Publishers, 2 nd Edition, 2019.
5	Wireless Communications: Principles and Practice”, Theodore Rappaport, Pearson, 2 nd Edition, 2016.
6	“Introduction to Embedded Systems”, Shibu K V, TATA Mc Graw Hill Edu, 2 nd Edition, 2016.

REFERENCE BOOKS:

1	Electronic Devices” Thomas L Floyd, Pearson, 9 th , 2012
2	D P Kothari, I J Nagrath, Mc Graw Hill, 2 nd , 2018

E Books / MOOCs/ NPTEL

1	https://nptel.ac.in/courses/117107095
2	https://nptel.ac.in/courses/117103063
3	https://www.coursera.org/learn/electronics/syllabus
4	https://www.coursera.org/learn/diode-pn-junction-metal-semiconductor-contact/specialization=semiconductor-devices
5	https://www.coursera.org/learn/transistor-field-effect-transistor-bipolarjunction-transistor/specialization=semiconductor-devices

ENVIRONMENTAL SCIENCE AND SUSTAINABILITY			
Course Code:	26CIV104	Course Type	BSC
Teaching Hours/Week (L: T: P: S)	2:0:0:0	Credits	02
Total Teaching Hours	30	CIE + SEE Marks	60+40
Prerequisite	*		
Teaching Department: Civil Engineering			
Course Objectives: This Course will enable students to:			
1.	To introduce the basic concepts of environment, ecosystems and biodiversity and emphasize on the biodiversity of India and its conservation		
2.	To impart knowledge on the causes, effects and control or prevention measures of environmental pollution and natural disasters		
3.	To facilitate the understanding of renewable and non-renewable resources, causes of their degradation and measures to preserve them.		
4.	To familiarize the concept of sustainable development goals.		
5.	To inculcate and embrace sustainability practices and develop a broader understanding on green materials, energy cycles.		
UNIT-I			
ENVIRONMENT AND BIODIVERSITY			05 Hours
Definition, scope and importance of environment – need for public awareness. Eco-system and Energy flow– ecological succession. Types of biodiversity: genetic, species and ecosystem diversity– values of biodiversity, India as a mega-diversity nation – hot-spots of biodiversity – threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – endangered and endemic species of India – conservation of biodiversity: In- situ and ex-situ.			
ENVIRONMENTAL POLLUTION			05 Hours
Causes, Effects and Preventive measures of Water, Soil, Air and Noise Pollutions. Solid, Hazardous and E-Waste management. Case studies on Occupational Health and Safety Management system (OHASMS). Environmental protection, Environmental protection acts.			
RENEWABLE SOURCES OF ENERGY			05 Hours
Energy management and conservation, New Energy Sources: Need of new sources. Different types new energy sources. Applications of- Bio energy, Hydrogen energy, Ocean energy resources, Tidal energy conversion. Concept, origin and power plants of geothermal energy.			
UNIT-II			
SUSTAINABILITY AND MANAGEMENT			07 Hours
Sustainability- concept, needs and challenges-environmental, social and economic aspects of sustainability- Sustainable Development Goals-targets, indicators and intervention areas-Protocols-Climate change- Global, Regional and local environmental issues and possible solutions-case studies. Concept of Carbon Credit, Carbon Footprint, water Foot print Environmental management in industry- A case study			
SUSTAINABILITY PRACTICES			08 Hours
Zero waste and R concept, Circular economy, LCA standards-ISO 14000 Series, Material Life cycle assessment, Environmental Impact Assessment. Sustainable habitat: green buildings, green materials, Energy efficiency, Sustainable transports. Sustainable energy: Non-conventional Sources, Energy Cycles- carbon cycle, emission and sequestration, Green Engineering: Sustainable urbanization- Socio- economic and technological change			

Course Outcomes: At the end of the course student will be able to																																																																																																																							
1.	To recognize and understand the functions of environment, ecosystems and biodiversity and their conservation.																																																																																																																						
2.	To identify the causes, effects of environmental pollution and natural disasters and contribute to the preventive measures in the society.																																																																																																																						
3.	To identify and apply the understanding of renewable and non-renewable resources and contribute to the sustainable measures to preserve them for future generations.																																																																																																																						
4.	To recognize the different goals of sustainable development and apply them for suitable technological advancement and societal development.																																																																																																																						
5.	To demonstrate the knowledge of sustainability practices and identify green materials, energy Cycles and the role of sustainable urbanization																																																																																																																						
Course Outcomes Mapping with Program Outcomes & PSO																																																																																																																							
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Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓																																																																																																											
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26CIV104.4	2	1				2	3	2				1	3																																																																																																										
26CIV104.5	1	1				2	3	2				1	3																																																																																																										
1: Low, 2: Medium, 3: High																																																																																																																							

TEXTBOOKS:															
1.	Anubha Kaushik and C. P. Kaushik 's —Perspectives in Environmental StudiesI, 6th Edition, New Age International Publishers ,2018.														
2.	Benny Joseph, _Environmental Science and Engineering ', Tata McGraw-Hill, New Delhi, 2016.														
3.	Gilbert McMasters, _Introduction to Environmental Engineering and Science', 2nd edition, Pearson Education, 2004														

REFERENCE BOOKS:															
1.	R.K. Trivedi, _Handbook of Environmental Laws, Rules, Guidelines, Compliances and Standards ', Vol. I and II, Enviro Media. 38 Edition 2010.														
2.	Cunningham, W.P. Cooper, T.H. Gorhani, _Environmental Encyclopedia', Jaico Publ., House, Mumbai, 2001.														
3.	Dharmendra S. Sengar, _Environmental law ', Prentice Hall of India PVT. LTD, New Delhi, 2007.														
4.	Rajagopalan, R, _Environmental Studies-From Crisis to Cure ', Oxford University Press, Third Edition, 2015.														
5.	Erach Bharucha —Textbook of Environmental Studies for Undergraduate Courses Orient Blackswan Pvt. Ltd. 2013														
6.	R.K. Trivedi, _Handbook of Environmental Laws, Rules, Guidelines, Compliances and Standards ', Vol. I and II, Enviro Media. 38. Edition 2010.														

UNIVERSAL HUMAN VALUES & PROFESSIONAL ETHICS

Course Code	26HSS102	Course Type	HSMC
Teaching Hours/Week (L:T:P:S)	02:0:0:0	Credits	02
Total Teaching Hours	30+0+0	CIE + SEE Marks	60+40

Teaching Department: Any

Course Objectives:

1	To enable the students to appreciate the 'values', "Skills" and 'Behaviour' with an appropriate understanding of the 'self' to attain sustained happiness and prosperity with right aspirations of life
2	To develop a holistic perspective among the students towards physical needs and prosperity of life
3	To develop a holistic approach and understanding the importance of co-existence and living in harmony ensuring mutually fulfilling interaction with society and nature

UNIT – I

10 Hours

Self-exploration – Content and Process; Natural Acceptance and Empirical Validation
Continuous Happiness and Prosperity – Understanding basic human aspirations; Mindfulness and awareness of right aspirations
Right Understanding, Relationships and Physical Facilities – Requirements for the fulfillment of aspirations with correct priority – Happiness and Prosperity

UNIT – II

10 Hours

Understanding Harmony in the 'Human being' and Harmony in 'Self' – Human being as a co-existence of sentient 'I' (Self) and 'Body'; Body as an instrument

UNIT – III

10 Hours

Understanding Harmony in Family and Society – Harmony in Human Relationships; Universal harmonious order in Family, Society and Universe (Journey from Family to World Family);
Personal and Professional Ethics;
Understanding Harmony in Nature and Existence – Existence as 'Co-existence'; Holistic Perception of Harmony at all levels of existence

Course Outcomes: At the end of the course, student will be able to

1.	have better self-exploration and understanding with a capacity to identify the priorities of life
2.	Generate sustainable solutions to problems with focus on human values and value-based living
3.	Understand and practice living in harmony, co-existence and natural acceptance at all levels
4.	Be ethically upright in their personal and professional practices

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓	
↓ Course Outcomes												1	2
26HSS102.1						3	2	2	1		2	-	-
26HSS102.2						3	2	2	1		2		

	26HSS102.3						3	2	2	1		2			
	26HSS102.4						3	2	1	1		2			
1: Low, 2: Medium, 3: High															

TEXT BOOK

- 1 'A Foundation Course in Human Values and Professional Ethics' by R.R. Gaur (Author), R.Sangal (Author), G.P. Bagaria (Author)

REFERENCE BOOKS

- 1 The power of Positive Thinking by Norman Vincent Peale
- 2 The power of Now by Eckhart Tolle
- 3 Mindfulness Yoga by Frank Jude Boccio

Websites,

Movies and Documents

- 1 Value Education websites, <http://uhv.ac.in>, <http://www.uptu.ac.in>
- 2 Story of Stuff, <http://www.storyofstuff.com>
- 3 Al Gore, *An Inconvenient Truth*, Paramount Classics, USA
- 4 Charlie Chaplin, *Modern Times*, United Artists, USA
- 5 IIT Delhi, *Modern Technology – the Untold Story*
- 6 *Gandhi A., Right Here Right Now*, Cyclewala Productions

BIOLOGY FOR ENGINEERS			
Course Code:	26BTY111	Course Type:	BSC
Teaching Hours/Week (L: T: P: J):	1:0:0:0	Credits:	01
Total Teaching Hours	15+0+0	CIE + SEE Marks:	60+40
Teaching Department: Biotechnology			

Course Objectives:	
1.	To learn the types of cells, biomolecules and life processes
2.	To know the applications inspired by nature in various streams and real-life scenarios.

UNIT-I	
INTRODUCTION FOR BIOLOGY FOR ENGINEERS	05 Hours

Why Biology for Engineers? Living cell Types & Properties: Prokaryotes - Bacteria, Viruses and Fungi, Eukaryotes - Plant and Animal Cells, Biomolecules, Life Processes at Cellular Level.

UNIT-II	
APPLICATIONS INSPIRED BY NATURE & REAL-LIFE SCENARIOS	10 Hours

Composites in Construction, Termite Mound architecture, Countercurrent heat exchangers, Design of Aeroplane, helicopter and submarine, Information Theory and Biology, SONAR, Medical Devices. Recent scenarios in Environment, Agriculture and Medical Technology.

Course Outcomes: At the end of the course students will be able to	
1.	Ascertain the importance of Biology and understand the basics of cell and life processes
2.	Draw the inspiration from nature in engineering design and visualize the application of biology in recent scenarios

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓	
↓ Course Outcomes												1	2
26BTY111.1	3	3	-	-	-	-	-	1	-	-	1	-	-
26BTY111.2	3	3	-	-	-	-	-	1	-	-	1		

1: Low, 2: Medium, 3: High

TEXTBOOKS:	
1	Suraishkumar, G.K. <i>Biology for Engineers</i> , Oxford University Press India, 2019.
2	Chakraborty, T, Akthar, N <i>Biology for Engineers</i> , PHI learning Print Book ISBN:9789391818142 eBook ISBN: 9789391818197

REFERENCE BOOKS:	
1.	Rao C.V., <i>Biology for Engineers</i> , 2021
2.	Raven, P. H. and Johnson, G. B. <i>Biology</i> . 4th Ed. WCB publishers, 2010.
3.	Ethier, R. S. and Simmons, C. A. <i>Introductory biomechanics- From cells to organisms</i> . Cambridge University Press, 2012

ENGINEERING VISUALIZATION			
Course Code:	26MEC122	Course Type	PCC
Teaching Hours/Week (L: T: P: S)	0:0:3:0	Credits	1
Total Teaching Hours	30	CIE + SEE Marks	60+00
Teaching Department: Mechanical Engineering			
Course Objectives:			
1.	To impart and inculcate understanding of the concept of orthographic projection and projection of plane surfaces and solids in different position in first angle projection system.		
2.	To draw the orthographic views of the simple machine parts and to draw the isometric projection of simple solids.		
UNIT-I			
Chapter 1: Orthographic Projection: Introduction to orthographic projection, Quadrants, principal planes, principal views, Difference between First angle and third angle projection, Dimensioning, Conventions employed for drawing. Projection of points and projection of lines(First quadrant only)			02 Hours
Chapter 2: Projection of plane surface: Triangle, Square, Rectangle, Pentagon, & Hexagon in simple position (Resting on HP with inclination to HP and VP, true length with true inclination only- No Beta angle)			06 Hours
UNIT-II			
Chapter 3: Projection of Solids: Prisms and Pyramids in simple position (Resting on HP with inclination to HP and VP, true length with true inclination only)			10 Hours
UNIT-III			
Chapter 5: Isometric projection: Isometric scale, Isometric dimensions, to draw Isometric views of simple solids and machine components using their orthographic projections.			06 Hours
Chapter 6: Pictorial View into Orthographic Views: Orthographic projection (First angle projection) of simple machine components using their pictorial view.			06 Hours
Course Outcomes: At the end of the course student will be able to			
1.	Draw the orthographic projections of a plane for a given position using Solid Edge software.		
2.	Draw the orthographic projections of a solids and simple machine parts for a given position using Solid Edge software.		
3.	Draw the orthographic views of the simple machine parts. Draw isometric projection of solid objects individually or in combination using Solid Edge software.		

Course Outcomes Mapping with Program Outcomes & PSO														
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓		
↓ Course Outcomes												1	2	
26MEC122.1	3	1							1	1		2	1	
26MEC122.2	3	1							1	1		2	1	
26MEC122.3	3	1							1	1		2	1	

1: Low, 2: Medium, 3: High

TEXTBOOKS:	
1.	Engineering Drawing by N. D. Bhat & V. M. Panchal, Pramod R. Ingle, 53 Ed. 2014, Charotar Publishing House, Gujarat.
2.	Engineering Drawing by K R Gopalakrishna, Subhas publishers, Bangalore , 32nd edition, 2012.
REFERENCE BOOKS:	
1.	A Primer on computer aided Engineering Drawing, Published by VTU, Belgaum, 8thedition, 2011.
2.	Engineering Drawing and Computer Graphics, Shah, Pearson, 2010
3.	Engineering Graphics, Agarwal & Agarwal, TMH, Second edition, 2013
4.	A Text book of Engineering Graphics And Drafting by P. S. GILL, 11th Ed.2009, S. K. Kataria & sons, ISBN- 8185749612, 9788185749617, New Delhi.

LINEAR ALGEBRA

Course Code:	26MAT106	Course Type:	BSC
Teaching Hours/Week (L-T-P):	3-1-0	Credits:	04
Total Teaching Hours:	45+15+0	CIE + SEE Marks:	60+40

Teaching Department: Mathematics

Course Objectives:

- The course will cover the concepts, methods to solve system of equations, evaluate Eigen values and Eigen vectors
- The course will impart knowledge of Linear algebra such as vector space, linear transformation, inner product space and matrix decomposition
- The course will also cover periodic function and Fourier analysis of periodic and aperiodic function.

UNIT-I

Matrices 9+3 Hours

Elementary transformation of a matrix, Echelon form and rank of a matrix. Consistency and solution of system of linear equations - Gauss elimination method. LU Decomposition method.

Eigen values and Eigen vectors of a square matrix, Characteristic and minimal polynomial, Rayleigh's power method.

UNIT-II

Vector Spaces 9+3 Hours

Vector spaces, subspaces, linear combination, spanning set, linearly dependent and independent vectors, basis and dimension, Change of basis, coordinates, row space, column space and null space.

UNIT-III

Linear Transformations 9+3 Hours

Linear transformations, algebra of linear transformations, representation of transformations by matrices, isomorphism, Range(image) and Null space(Kernel) of a linear transformation. Rank – nullity theorem. Inner products, orthogonal sets of projections, Gram-Schmidt's orthogonalization process.

UNIT-IV

Matrix Decompositions 9+3 Hours

Diagonalization, Jordan Canonical form, Quadratic forms, QR-factorization, least-squares problems, and singular value decomposition.

UNIT-V

Fourier series and Integral Transforms 9+3 Hours

Euler's formulae, Dirichlet's conditions for Fourier series expansion.

Fourier Transforms: Definition, Complex Fourier transforms, Cosine and Sine transforms, Inverse Fourier transforms.

Laplace Transforms: Definition, Laplace Transform of elementary functions, First shifting property, Transforms of periodic function, unit step function, unit impulse function, derivatives, integrals, multiplication by t^n , Division by t (only statements of properties and its problems). Inverse Laplace transforms by the method of partial fractions.

Course Outcomes: At the end of the course student will be able to

- 1 Apply concept of Eigen values to engineering problems and solve system of equation numerically
- 2 Construct vector spaces and analyse the properties like basis and dimensions
- 3 Analyse problems adopting concepts of Linear transformation, Orthogonality and inner products
- 4 Apply concepts of decomposition and canonical forms to engineering problems
- 5 Apply Fourier analysis to periodic and aperiodic functions

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓		
↓ Course Outcomes														
26MAT106.1	3	1	-	-	-	-	-	-	-	-	-	-	-	-
26MAT106.2	2	3	-	-	-	-	-	-	-	-	-	-	-	-
26MAT106.3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
26MAT106.4	3	2	-	-	-	-	-	-	-	-	-	-	-	-
26MAT106.5	2	2	-	-	-	-	-	-	-	-	-	-	-	-

1: Low, 2: Medium, 3: High

TEXTBOOKS:

- 1 Seymore Lipschutz, Marc Lipson, “Linear Algebra” 3rd edition, Schaum series, Tata McGraw Hill, 2011
- 2 Erwin Kreyszig, “Advanced Engineering Mathematics”, John Wiley and Sons, 10th Edition (Reprint), 2016
- 3 B.S. Grewal, “Higher Engineering Mathematics”, Khanna Publications, 43rd Edition, 2015.

REFERENCE BOOKS:

- 1 David C. Lay, “Linear Algebra and Its Applications”, Pearson Education, Inc., 5th edition, 2016.
- 2 Gilbert Strang, “Introduction to Linear Algebra”, Wellesley-Cambridge Press, 5th edition, 2016
- 3 J.W. Brown and R.V. Churchill, “Fourier Series and Boundary Value Problems”, McGraw-Hill Education, 8th Edition, 2011
- 4 Gilbert Strang “Linear Algebra for everyone” 2020, PHI
- 5 J R Hanna, J H Rowland, Fourier Series, Transforms and Boundary Value Problems, Dover, 2018

E Books / MOOCs/ NPTEL

- 1 <https://nptel.ac.in/courses/111101115>
- 2 <https://archive.nptel.ac.in/courses/111/106/111106135/>
- 3 https://onlinecourses.nptel.ac.in/noc24_ma21/preview

MATERIALS CHEMISTRY FOR DEVICES AND E-WASTE MANAGEMENT

(EE/EC/ACT/VLSI and All IT branches)

Course Code:	26CHY104	Course Type:	BSC
Teaching Hours/Week (L: T:P: S):	3:0:2:0	Credits:	04
Total Teaching Hours:	45+0+30	CIE + SEE Marks:	60+40

Teaching Department: Chemistry

Course Learning Objectives:

1.	To explain the principles and applications of various electrodes and energy systems, including batteries and solar cells, in sustainable technologies
2.	To explore corrosion mechanisms and control methods, alongside metal finishing processes such as electroplating and electroless plating
3.	To understand the principles and applications of sensors, memory devices, and advanced display systems in modern engineering
4.	To examine the synthesis, properties, and applications of polymers and nanomaterials in engineering
5.	To learn the principles of green chemistry and e-waste management, emphasizing sustainable practices in engineering applications.

UNIT-I

ELECTRODES AND ENERGY SYSTEMS **09 Hours**

Electrodes: Introduction, types of electrodes; Reference electrode - introduction, Calomel electrode – construction, working, advantages and applications; Concentration cell – definition, construction, working, and numerical problems; Ion selective electrode – introduction, glass electrode - construction, working and advantages of glass electrode, determination of pH using glass electrode, Concentration cell – definition, construction, working, and numerical problems;

Energy Systems: Introduction to batteries; classification of batteries; construction, working and applications of Li-ion battery; Flow battery – introduction, construction, working and applications of vanadium redox flow battery; Construction, working and applications of solar photovoltaic cell, advantages and disadvantages; synthesis of solar grade silicon by union carbide method

UNIT II

CORROSION SCIENCE AND METAL FINISHING **09 Hours**

Corrosion Science: Introduction; electrochemical theory of corrosion (taking Fe as example), corrosion types – differential metal corrosion, differential aeration corrosion, pitting corrosion; factors affecting corrosion – primary and secondary factors; corrosion penetration rate – numerical problems; corrosion control – metal coating (galvanization and tinning), anodization, sacrificial anode method,

Metal Finishing: Introduction; electroplating – definition, principle, factors affecting electroplating, electroplating of chromium (hard chromium and decorative chromium); electroless plating – definition, advantages, application of electroless plating of copper in the manufacture of PCB; comparison between electroplating and electroless plating

UNIT-III

ANALYTICAL TECHNIQUES, MEMORY DEVICES AND DISPLAY SYSTEMS **09 Hours**

Analytical technique and sensors: Conductometry- principle, instrumentation, conductometric titration of strong acid and acid mixture; Potentiometry- principle, instrumentation and application of potentiometry in the estimation of iron; determination of pKa of weak acid, Colorimetry- principle (Beer Lambert's law), instrumentation and application in determination of Cu/Fe; **Electrochemical sensors** – introduction, construction and working of electrochemical sensors for detection oxides of sulphur (Industry), diclofenac (medicine); disposable sensors and their advantages

Memory Devices: Introduction; basic concepts of electronic memory; classification of electronic memory devices; types of organic memory devices (organic molecules, polymeric materials, organic-inorganic hybrid materials) Display Systems: Liquid crystals (LCs) – introduction; classification of LCs and types of mesophases; electro-optic effect; Liquid Crystal Display (LCD) unit – construction and application in numerical display; characteristics, working and applications of organic light emitting diode (OLED) and quantum light emitting diode (QLED).	
UNIT IV	
POLYMERS AND NANOMATERIALS	09 Hours
Polymers: Introduction; classification based on origin, structure, methods of polymerization, response to heat; Molecular weight of polymers: number average molecular weight, weight average molecular weight and poly dispersion index (PDI) - numerical problems;; Polymer composites – introduction, synthesis, properties and applications of carbon fiber; Conducting polymers – introduction, conducting mechanism of polyacetylene by oxidative and reductive doping, applications of conducting polymers. Nanomaterials: Introduction; classification based on dimension; catalytic and thermal properties of nanoparticles; synthesis of nanomaterials by top-down approach (photolithography) and bottom-up approach (sol-gel method); synthesis by CVD technique	
UNIT-V	
E-WASTE MANAGEMENT	09 Hours
E-waste management: Introduction; sources of e-waste; composition, characteristics and need of e-waste management; health hazards due to exposure to e-waste; heavy metals present in the E-waste and their ill effect on health and environment. Extraction of metals from e-waste- Pretreatment, Pyrometallurgy- smelting, combustion, incineration and pyrolysis. Hydrometallurgy- Use of different leaching agents in metal extraction, Metal recovery- Use of solvents, adsorption, ion exchange and electrodeposition, extraction of gold and copper from e-waste. . Role of stakeholders in environmental management of e-waste (producers, consumers, recyclers, and statutory bodies).	
List of Experiments (Any ten)	
1.	Determination of strength of an acid in Pb-acid battery.
2.	Determination of Total Hardness of a sample of water using disodium salt of EDTA.
3.	Estimation of iron in TMT bar by diphenylamine/external indicator method.
4.	Conductometric estimation of strong acid/weak acid with standard NaOH solution.
5.	Conductometric estimation of acid mixture with standard NaOH solution.
6.	Potentiometric estimation of FAS using standard $K_2Cr_2O_7$ solution.
7.	Determination of pKa of vinegar using pH sensor (Glass electrode).
8.	Determination of the viscosity coefficient of a given liquid using Ostwald's viscometer.
9.	Estimation of Copper present in electroplating effluent by optical sensor (colorimetry).
10.	Colorimetric determination of iron.
11.	Estimation of Sodium present in soil/effluent sample using a flame photometer.
12.	Synthesis of biodiesel (Demonstration).
13.	Synthesis of Iron-oxide Nanoparticles (Demonstration).
14.	Synthesis of polyurethane (Demonstration).

Course Outcomes: At the end of the course students will be able to

1	Apply the principles, types, and applications of various electrodes and energy systems, such as batteries and solar cells, in promoting sustainability in engineering
2	Apply corrosion control techniques, including metal finishing methods like electroplating and electroless plating, in real-world engineering contexts.
3	Analyze the principles and evaluate the applications of sensors, memory devices, and display systems in modern engineering technologies.
4	Analyze the synthesis, properties, and practical applications of polymers and nanomaterials in various engineering fields.
5	Apply the principles of green chemistry and assess e-waste management strategies, focusing on sustainability and environmental impact in engineering practices.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓		
↓ Course Outcomes														
26CHY104-1.1	3	1												
26CHY104-1.2	3	2												
26CHY104-1.3	3	2												
26CHY104-1.4	2	1												
26CHY104-1.5	2	1												

1: Low, 2: Medium, 3: High

TEXTBOOKS:

1	Engineering Chemistry by P.C. Jain & Monica Jain., Dhanpat Rai Publications, New Delhi.2015.
2	A Textbook of Engineering Chemistry, R. V. Gadag and Nityananda Shetty, I. K. International Publishing house. 2nd Edition, 2016.
3	A Textbook of Engineering Chemistry, SS Dara &Dr. SS Umare, S. Chand & Company Ltd., 12th Edition, 2011.

REFERENCE BOOKS:

1	Wiley Engineering Chemistry, Wiley India Pvt. Ltd. New Delhi, 2013-2 nd Edition.
2	Engineering Chemistry, Satya Prakash & Manisha Agrawal, Khanna Book Publishing, Delhi.
3	Essentials of Physical Chemistry, Bahl & Tuli, S. Chand Publishing.
4	Applied Chemistry, Sunita Rattan, Kataria 5. Engineering Chemistry, Baskar, Wiley.
5	Engineering Chemistry – I, D. Grouer Krishana, Vikas Publishing.
6	Textbook of Polymer Science, F.W. Billmeyer, John Wiley & Sons, 4 th Edition, 1999.
7	Nanotechnology A Chemical Approach to Nanomaterials, G.A. Ozin & A.C. Arsenault, RSC Publishing, 2005.
8	Linden's Handbook of Batteries, Kirby W. Beard, Fifth Edition, Mc GrawHill, 2019.
9	OLED Display Fundamentals and Applications, Takatoshi Tsujimura, Wiley–Blackwell, 2012
10	Supercapacitors: Materials, Systems, and Applications, Max Lu, Francois Beguin, Elzbieta Frackowiak, Wiley-VCH; 1 st edition, 2013.
11	Handbook on Electroplating with Manufacture of Electro-chemicals”, ASIAPACIFIC BUSINESS PRESS Inc., 2017. Dr. H. Panda.
12	Laboratory manual in Engineering Chemistry Sudharani, Dhanpat Rai Publishing Company, New Delhi.

13	Expanding the Vision of Sensor Materials. National Research Council 1995, Washington, DC: The National Academies Press. doi:10.17226/4782.
14	Engineering Chemistry, Edited by Dr. Mahesh B and Dr. Roopa Shree B, Sunstar Publisher, Bengaluru, ISBN978-93-85155-70-3, 2022.
15	High-Performance Metallic Materials for Cost-Sensitive Applications, F.H. Froes, et al. John Wiley & Sons, 2010.
16	Instrumental Methods of Analysis, Dr. K. R. Mahadik, and Dr. L. Satyanarayana, Nirali Prakashan, 2020.
17	Principles of Instrumental Analysis, Douglas A. Skoog, F. James Holler, Stanley R. Crouch Seventh Edition, Cengage Learning, 2020.
18	Polymer Science, VR Gowariker, N V Viswanathan, Jayadev, Sreedhar, New Age Int. Publishers, 4th Edition, 2021.
19	Nanostructure materials and nanotechnology, Hari Singh, Nalwa, academic press, 1st edition, 2002.
20	“Engineering Chemistry”, O. G. Palanna, Tata Mc Graw Hill Education Pvt. Ltd. New Delhi, Fourth Reprint, 2015.
E-Books / MOOCs/ NPTEL	
1	http://libgen.rs/ • https://nptel.ac.in/downloads/122101001/
2	https://nptel.ac.in/courses/104/103/104103019/ • https://ndl.iitkgp.ac.in/ .
3	https://www.youtube.com/watch?v=faESCxAWR9k

APPLIED DIGITAL LOGIC DESIGN

Course Code:	26ECE101	Course Type:	ESC
Teaching Hours/Week (L: T:P: S):	2-0-2-0	Credits:	3
Total Teaching Hours:	30+15	CIE + SEE Marks:	60+40

Teaching Department: Electronics and Communication Engineering/Allied Branches

Course Objectives:

1	To enable students to understand and simulate fundamental concepts of digital electronics including number systems, binary arithmetic, logic gates, and Boolean algebra.
2	To introduce methods for simplifying Boolean expressions and verifying them through simulation using basic logic gates.
3	To equip students with the knowledge and simulation skills to analyze and design combinational logic circuits.
4	To provide students with an understanding of flip-flops and their simulation for sequential circuit behavior.
5	To enable students to apply flip-flops in the design and simulation of sequential circuits such as shift registers and counters.

UNIT-I

Fundamentals of Digital Design	06 Hours
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Difference between Analog and Digital Signals, Number Systems: Decimal, Binary, Octal and Hexadecimal. Binary Addition and Subtraction, Digital Logic Gates, Boolean Algebra, Boolean Functions: Canonical Forms, Completely and Incompletely Specified Functions

UNIT-II

Simplification of Boolean expressions	06 Hours
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Simplification of Boolean Functions using Boolean Algebra, Karnaugh Maps, Quine-McCluskey (up to four variables), Realization of Boolean functions using Basic Gates

UNIT-III

Combinational Logic Circuits	06 Hours
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Introduction to Combinational Logic Circuits, Half/Full Adders/Subtractors, Parallel Adders/Subtractors, Binary Comparators, Decoders, Encoders, Multiplexers.

UNIT-IV

Sequential Logic Circuits	06 Hours
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Basic Bistable Element, SR Flip-Flop, D Flip Flop, JK Flip Flop, T Flip Flop, Master Slave JK Flip Flop, Characteristic Equations, Conversion of Flip Flops.

UNIT-V

Applications of Flip Flops	06 Hours
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Design of Shift Register using D- flip flop, Design of Counters: Asynchronous counters using T-flip flop, Synchronous Counters using D-flip flop and T Flip Flop.

List of Experiments

(Use this section wherever applicable, else delete this section)

1.	Introduction to Simulation tool
2.	Simulation of Basic Logic Gates and Binary Arithmetic Operations
3.	Simulation and Verification of Simplified Boolean Functions using K-Map and Logic Gates
4.	Simulation and verification of a Half Adder and Full adder
5.	Simulation and verification of a Half subtractor and Full subtractor
6.	Simulation and verification of a Binary Adder and Multiplexer Circuit
7.	Simulation and verification of D,T,SR and JK Flip-Flops
8.	Simulation of Flip-Flops and Conversion Between Flip-Flop Types
9.	Simulation of 4-bit Shift Register and Asynchronous Counters
10.	Simulation of 4-bit Shift Register and Synchronous Counters

Course Outcomes: At the end of the course student will be able to:

1	Demonstrate binary operations, convert between number systems, and simulate basic digital logic circuits using Boolean functions.
2	Simplify Boolean functions using Boolean algebra, Karnaugh maps, and the Quine-McCluskey method, and simulate the implementation using basic logic gates.
3	Design and simulate combinational circuits such as adders, subtractors, comparators, decoders, encoders, and multiplexers.
4	Describe, simulate, and analyze various flip-flops, perform conversions, and determine their characteristic equations.
5	Design and simulate shift registers and both asynchronous and synchronous counters using appropriate flip-flops.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓	
↓ Course Outcomes												1	2
26ECE101.1	3	2	2		3							3	2
26ECE101.2	3	2	3		3							3	2
26ECE101.3	3	2	3		3							3	3
26ECE101.4	3	2	2		3							3	2
26ECE101.5	3	2	3		3							3	3

1: Low, 2: Medium, 3: High

TEXTBOOKS:

1	Digital Principles and Design, Donald D. Givone, McGraw Hill, ,2002
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REFERENCE BOOKS:

1	Digital Design, Morris Mano, Prentice Hall of India, 3rd Edition
2	Digital Logic Applications and Design, John M Yarbrough, Thomson Learning,2001
3	D. P. Kothari and J. S Dhillon, Pearson,2016
4	Fundamentals of Logic Design, Charles H Roth, Cengage Learning

E Books / MOOCs/ NPTEL

1	https://nptel.ac.in/courses/117106086
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ELEMENTS OF ELECTRICAL ENGINEERING			
Course Code	26EEE102	Course Type	ESC
Hours/Week (L:T:P:S)	3:0:0:0	Credits	3
Total Teaching Hours	45+0+0	CIE + SEE Marks	60+40
PREREQUISITE:			
Course Objectives			
1.	To understand fundamental electrical laws and apply graph theory to analyze electrical circuits.		
2.	To analyze single-phase AC circuits and calculate power and power factor.		
3.	To understand conventional and non-conventional power generation and basics of power transmission.		
4.	To explain power distribution systems and principles of DC-DC converters.		
5.	To understand the fundamentals of electrical machines, protection devices and power supply systems		
Course Contents			
UNIT – 1			9 Hours
Graph Theory in Electrical Circuits			
Ohm’s law, Kirchhoff Laws (KVL, KCL). Graph Representation: Nodes, Branches, Linear graph of a network, Tie set and Cut-set Matrix, Incidence Matrix, Analysis of resistive network using cut-set and tie-set			
UNIT – 2			9 Hours
Analysis of AC Circuits			
Instantaneous values, RMS values, Average values, Form factor, Peak factor. Single-phase AC Circuits: R, L, C, RL, RC, RLC Circuits, Power Factor, Power			
UNIT - 3			9 Hours
Introduction to Power System			
Energy Sources (Conventional -Hydro, Nuclear, Thermal, Hydroelectric power plants & Non-conventional- Solar, Wind, Tidal), Basic power plant components, Voltage levels in generation Transmission System Overview (HVAC & HVDC), Transmission Voltage Levels (Low, Medium, High, Extra High Voltage)			
Overview of Power Distribution			
Primary & Secondary Distribution, Distribution Voltage Levels (415V, 230V)			
UNIT - 4			9 Hours
Introduction to DC-DC Conversion			
Working principle and application of Buck Converter, Boost Converter, Uninterruptible Power Supply (UPS)- Block Diagram & Working			
Electric Vehicles (EV)			
Fundamentals, Block diagram of EV and its components, Motors used in EV			
UNIT - 5			9 Hours
Overview of Electrical Machines, Protective Devices			
Magnetic Circuits Concepts, dot convention, Transformers -Working Principle, Types, Applications Electrical Machines: DC Motor, Stepper Motor - working principle and applications Protective devices-Fuse, MCB Earthing – Necessity, Types, Ground and neutral concept			

Course Outcomes																																																																																															
At the end of the course student will be able to																																																																																															
1.	Analyze resistive electrical circuits using graph theory and Kirchhoff laws.																																																																																														
2.	Solve single-phase AC circuits and evaluate power quantities.																																																																																														
3.	Identify and differentiate among various energy sources and explain the structure of the power system.																																																																																														
4.	Explain the working of DC-DC converters and components used in Electric Vehicle.																																																																																														
5.	Describe the working principles of transformers, electrical machines, protective devices.																																																																																														
Course Outcomes Mapping with Program Outcomes & PSO																																																																																															
<table><tr><th>Program Outcomes→</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th><th>10</th><th>11</th></tr><tr><th>↓ Course Outcomes</th><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>26EEE102.1</td><td>3</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr><tr><td>26EEE102.2</td><td>3</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr><tr><td>26EEE102.3</td><td>3</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr><tr><td>26EEE102.4</td><td>3</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr><tr><td>26EEE102.5</td><td>3</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr></table>												Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	↓ Course Outcomes												26EEE102.1	3	2						1	1			26EEE102.2	3	2						1	1			26EEE102.3	3	2						1	1			26EEE102.4	3	2						1	1			26EEE102.5	3	2						1	1		
Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11																																																																																				
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26EEE102.5	3	2						1	1																																																																																						
1: Low, 2: Medium, 3: High																																																																																															
TEXTBOOKS																																																																																															
1. Electrical Technology, Hughes Edward, Pearson Education Publications, 10 th Ed, 2010																																																																																															
2. Electrical Power Systems, C.L.Wadhwa, New Age International, 7 th Ed, 2016																																																																																															
3. Power Electronics: Circuits, Devices and Applications, Muhammad H. Rashid, 4 th Editon, 2017																																																																																															
REFERENCE BOOKS																																																																																															
1. Basic Electrical Engineering, D.P. Kothari and I.J. Nagrathm, McGraw Hill Education, 4 th Ed, 2019																																																																																															
2. 500 Solutions of Problems in Electrical Engineering Part I, Parker Smith, CBS, 2003																																																																																															

INTRODUCTION TO PYTHON PROGRAMMING

Course Code:	26CSE102	Course Type:	PLC
Teaching Hours/Week (L:T:P:J:SL)	2:0:2:0:1	Credits:	3
Total Teaching Hours:	30:0:30:0:15	CIE + SEE Marks:	60+40

Teaching Department: Any

Course Objectives:

1	Understand basic Python programming constructs.
2	Develop modular programs using Python functions.
3	Utilize Python data structures.
4	Perform file operations and apply object-oriented programming.
5	Handle exceptions and visualize data.

UNIT-I

Introduction to Python Programming	6 Hours
Python Concepts: Introduction to Python, Variables, Keywords, Identifiers, Literals, Comments, Operators	
Control and Looping statements: if, if-else, elif, nested if, for, while, break, continue, pass	

UNIT-II

Functions and Modular Programming	6 Hours
Design with functions: Functions Overview, Arguments and return values, Formal vs Actual arguments, Named arguments, Recursive Functions, Lambda Functions, Modules	

UNIT-III

Data Structures in Python	6 Hours
Data Structures: Strings and basic operations, Lists, Tuples, Sets, Dictionaries: creation and manipulation	

UNIT-IV

File Handling and Introduction to Object-Oriented Concepts	6 Hours
Files: Create a file, read and write operations with text and CSV files	
Object Oriented Concepts: Class, Object	

UNIT-V

Exception Handling and Data Visualization	6 Hours
Exception handling: Introduction to Exceptions and Errors, Handling exceptions using try-except-else-finally	
Introduction to Pandas: Data analysis and manipulation using Pandas	
Data Visualization using Matplotlib: Plotting different kinds of graphs using Matplotlib	

Suggested list of Experiments

1.	Experiments related to basic operation, data types, and variables.
2.	Experiments related to control and looping statements.
3.	Experiments on writing functions and parameter passing.
4.	Experiments related to the operations of lists, tuples, and dictionaries.
5.	Experiments related to working with strings.
6.	Experiments related to file handling.
7.	Experiments related to classes and objects.
8.	Experiments related to exception handling.
9.	Experiments related to data analysis and manipulation using Pandas.
10.	Experiments related to plotting graphs using Matplotlib.

Course Outcomes: At the end of the course student will be able to:

1	Apply core Python programming concepts such as variables, operators, and control flow to develop interactive programs.
2	Design and implement reusable code using functions and modules to solve real-world computational problems.
3	Manipulate and process data effectively using built-in Python data structures like strings, lists, tuples, sets, and dictionaries.
4	Demonstrate the ability to manage files and implement object-oriented principles such as classes and objects in Python.
5	Develop robust Python applications with exception handling and use data analysis and visualization tools like Pandas and Matplotlib.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11
↓ Course Outcomes											
26CSE102.1	3	2	-	-	1	-	1	-	-	-	3
26CSE102.2	3	2	-	-	1	-	1	-	-	-	3
26CSE102.3	3	2	-	-	1	-	1	-	-	-	3
26CSE102.4	3	2	-	-	1	-	1	-	-	-	3
26CSE102.5	3	2	-	-	1	-	1	-	-	-	3

1: Low, 2: Medium, 3: High

TEXTBOOKS:

1	Kenneth A. Lambert, “—The Fundamentals of Python: First Programs”, Cengage Learning, 3 rd Edition, 2023.
2	Yashwanth Kanetkar, Adithya Kanetkar, —Let us Python, 6th Edition, 2023.
3	Al Sweigart, “Automate the Boring Stuff with Python”, 1 st Edition, No Starch Press, 2015.

REFERENCE BOOKS:

1	Mark Summerfield, “Programming in Python 3 - A Complete Introduction to the Python Language”, Second Edition, Addison-Wesley, 2009.
2	Y. Daniel Liang, “Introduction to Programming Using Python”, Pearson, 2018.

E Books / MOOCs/ NPTEL

1	https://assets.openstax.org/oscms-prodcms/media/documents/Introduction_to_Python_Programming_-_WEB.pdf
2	https://baou.edu.in/assets/pdf/BSCIT_405_slm.pdf
3	https://www.iimchyderabad.com/Material/introduction-to-python-programming.pdf
4	https://cfm.ehu.es/ricardo/docs/python/Learning_Python.pdf

COMMUNICATIVE ENGLISH			
Course Code	26HSS131	Course Type	AEC
Teaching Hours/Week (L:T:P:S)	1:0:2:0	Credits	02
Total Teaching Hours	15+0+30	CIE + SEE Marks	60+40
Teaching Department: Humanities			
Course Objectives:			
1.	Develop proficiency in phonetics, pronunciation, and effective oral communication.		
2.	Strengthen foundational grammar and language skills.		
3.	Improve academic writing skills for various contexts by appreciating texts.		
4.	Enhance comprehension and analytical thinking through the skill of writing		
Unit – I			
Phonetics & Communication			15 Hours
• Introduction to Phonetics, IPA, stress, rhythm, and intonation. • Rhythm, Intonation, Past Tense and Plural Pronunciation, Awareness of Accents • Fundamentals of Communication, Barriers, Strategies, Requests and Permissions. • Verbal and Non-Verbal Communication, Telephone etiquette, greetings, requests. • Short Formal Speech			
Practical Activities:			
• Phonetic drills, speech exercises, interactive speaking activities. • Communication and telephone conversation simulations			
Unit – II			
Language Skills & Grammar			15 Hours
• Texts – Prose & Poetry • Grammar Essentials: Prepositions, Articles, Sentence structure. • Language Skills: Sentence construction, comprehension, and structured writing techniques. • Comprehension			
Practical Activities:			
• Grammar exercises, sentence correction, editing tasks. • Reading comprehension exercises and discussion-based learning			
Unit – III			
Writing Skills			15 Hours
• Cohesive Devices: Conjunctions, linkers, and sequence of ideas in writing. • Writing Skills: Paragraph development, linkers, structured writing. • Argumentation and Refutation • Application Letter			
Practical Activities:			
• Writing focused on clarity, coherence, and structure.			

Course Outcomes: At the end of the course students will be able to

1.	Identify the nuances of phonetics, intonation, and pronunciation to appreciate and incorporate Received Pronunciation
2.	Interpret and assess nuances of oral communication skills and non-verbal communication for professional usage
3.	Identify, interpret, and describe the critical ideas, values, and themes to appreciate literary pieces for its language and social interpretations
4.	Implement English vocabulary in both personal and professional contexts, enhancing language proficiency while also honing effective writing skills adaptable to various forms of written communication.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓	
↓ Course Outcomes												1	2
26HSS131.1	1	1					2		2		3		
26HSS131.2	2					2			3		3		
26HSS131.3		2				3	2		3		3		
26HSS131.4		2				2		2	2		2		

1: Low, 2: Medium, 3: High

TEXT BOOK

1. Tidings, Victor, R. et al. (2022). A Textbook of English Language & Communication Skills.

REFERENCES

1. Jones, D (2011). English Pronunciation Dictionary.
2. Woods (2016). A Remedial English Grammar for Foreign Students.
3. Kumar, S. (2011). Communication Skills. Oxford University Press.
4. CIEFL, Hyderabad (Year). Exercises in Spoken English Part I. Oxford University Press.
5. CIEFL, Hyderabad (Year). Exercises in Spoken English Part II. Oxford University Press.
6. CIEFL, Hyderabad (Year). Exercises in Spoken English Part III. Oxford University Press.
7. Zinsser, W. (1976). On Writing Well.
8. Swan (2014). Practical English Usage. Oxford University Press.
9. Lyons, L. H. (2006). Study Writing. Cambridge University Press.
10. Subhashini, Victor, R. et al. (2022). A Textbook of English Language & Communication Skills.

E Resources

1. <https://www.macmillandictionary.com/dictionary/british/>

MATHEMATICS WITH MATLAB

Course Code:	26MAT107	Course Type:	BSC
Teaching Hours/Week (L-T-P):	0-0-2	Credits:	01
Total Teaching Hours:	0+0+26	CIE + SEE Marks:	60+00

Teaching Department: Mathematics

Course Objectives:

This course will enable the students to learn to use MATLAB for basic operations, array handling, plotting, and solving problems using symbolic and numerical methods.

List of Experiments

1. Introduction to MATLAB: Basic Operators: Arithmetic, Logical and Relational Operators. Elementary math functions such as algebraic, trigonometric, logarithmic, exponential functions, Conditions and Loops.
2. Symbolic Computation, plotting curves, surfaces and vector fields.
3. Computation of angle between (a) radius vector and tangent (b) two curves
4. Computation of radius of curvature
5. Taylor's/ Maclaurin's series expansion of a function of a single variable.
6. Differentiation of composite and computation of partial derivatives.
7. Computation of Jacobian.
8. Computation of extreme values of multivariate functions.
9. Solution of first-order ordinary differential equation and second and higher-order linear differential equation with constant coefficients
10. Evaluation of double/triple integrals with constant/variable limits.

Course Outcomes: At the end of the course student will be able to

1	Introduction to MATLAB: Basic Operators: Arithmetic, Logical and Relational Operators. Elementary math functions such as algebraic, trigonometric, logarithmic, exponential functions, Conditions and Loops.
2	Symbolic Computation, plotting curves, surfaces and vector fields.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓		
↓ Course Outcomes														
26MAT107.1	3	2	-	-	-	-	-	-	-	-	-	-	-	-
26MAT107.2	2	2	-	-	-	-	-	-	-	-	-	-	-	-
26MAT107.3	3	2	-	-	-	-	-	-	-	-	-	-	-	-
26MAT107.4	3	2	-	-	-	-	-	-	-	-	-	-	-	-
26MAT107.5	3	2	-	-	-	-	-	-	-	-	-	-	-	-

1: Low, 2: Medium, 3: High

REFERENCE BOOKS:	
1	Rudra Pratap, “MATLAB”, OXFORD University press, 2010
2	Dorothy C. Attaway Ph.D., A practical introduction to prog. And problem solving, 5th edition
E Books / MOOCs/ NPTEL	
1	https://www.mathworks.com › matlab › matlab_prog
2	https://www.coursera.org/specializations/mathematics-engineers
3	https://www.coursera.org/specializations/matlab-programming-engineers-scientists
4	https://www.coursera.org/learn/matlab

CONSTITUTION OF INDIA & GLOBAL CITIZENSHIP

Course Code	26HSS101	Course Type	HSMC
Teaching Hours/Week (L: T:P:S)	01:0:0:0	Credits	01
Total Teaching Hours	15+0+0	CIE + SEE Marks	60 + 00

Teaching Department: Humanities

Course Objectives:

1.	To Understand the Obligations, Responsibilities, Privileges and Rights, Duties and the Role that they have to play in deciding the Administrative Machinery of the nation as citizens.
2.	To empower learners to become aware of and understand global issues and to become active promoters of more peaceful, tolerant, inclusive, secure, and sustainable societies.

UNIT - I

Evolution of the Indian Constitution	05 Hours
Cabinet Mission Plan of 1946 – Partition Plan of 1947 and Indian Independence Act 1947, Constituent Assembly and Committees – Drafting, Enactment and Enforcement of Indian Constitution, Basic structure of Indian Constitution, Salient Features of Indian Constitution, Fundamental Rights, Fundamental Duties.	
Separation of Powers (Organs of the Constitution)	03 Hours
Executive: Union and State Legislature: Union and State Indian Judicial System	

UNIT - II

Structure of Government	04 Hours
Union: Prime Minister, Council of Ministers State: Chief Minister, Council of Ministers Local Self-Governance: Panchayat Raj Institutions, Urban Governance	
Global Citizenship	03 Hours
Meaning of Global Citizenship – ‘VasudhaivaKutumbakam’, Types of Global Citizenship, Globalization and Interconnectedness, Characteristics/Attributes of a Global Citizen, Rights of Global Citizenship, Responsibilities of a Global Citizen, Ethical Dimensions and Sustainability practices.	

Course Outcomes: At the end of the course, student will be able to

1.	Analyze the legalities and related issues of Indian Constitution as a fundamental law of the nation and the rights and duties of Indian Citizen.
2.	Become a responsible citizen being interconnected; has knowledge about his/her role in the community, State and the country; has a role in making the world a better place to live by applying critical thinking and ethical reasoning.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	PSO↓	
↓ Course Outcomes												1	2

26HSS101.1	-	-	-	-	-	3	2	-	-	-	2	-	-
26HSS101.2	-	-	-	-	-	3	2	1	1		2	-	-
1: Low, 2: Medium, 3: High													

REFERENCE BOOKS

1	Basu, D.D., "Introduction to the Constitution of India", Twentieth Edition, Reprint 2011. LexisNexis Butterworths Wadhwa, Nagpur, Haryana, India, 2011.
2	Pylee, M.V. "Introduction to Constitution of India", Fourth Revised Edition. Vikas Publishing House Pvt. Ltd., New Delhi, 2005.
3	Sharma, B.K. "Introduction to Constitution of India", Second Edition. Prentice Hall of India Pvt. Ltd., New Delhi, 2004.
4	Venkatesh, B.R. & Merunandan, K.B., "An Introduction to Constitution of India and Professional Ethics", Second Edition. Merugu Publications, Bangalore, 2007.
5	United Nations. "The Sustainable Development Goals Report."
6	UNESCO. (2015). "Global Citizenship Education: Topics and Learning Objectives."
7	Amartya Sen. (1999). "Development as Freedom."
8	Martha Nussbaum. (2011). "Creating Capabilities: The Human Development Approach."
9	Martha Nussbaum, <i>Cultivating Humanity: A Classical Defense of Reform in Liberal Education</i>

E Resources

1	NPTEL course: "Constitution of India". http://nptel.ac.in/courses/109104032/
2	"Constitution of India" eBook. https://pothi.com/pothi/book/ebook-ministry-law-and-justice-constitution-india .
3	iasplanner.blogspot.com/2010/11/free-ebook-download-constitution-of.html
4	www.iasabhiyan.com
5	Samvidhaan, Documentary by Prasaar Bharathi
6	http://en.unesco.org/gced
7	https://www.theglobalcitizensinitiative.org/the-rights-and-responsibilities-of-global-citizenship/
8	https://www.un.org/en/academic-impact/global-citizenship
9	UN Sustainable Development Goals: https://sdgs.un.org/
10	UNESCO Global Citizenship Education: https://en.unesco.org/themes/gced

KNOWING YOURSELF																
Course Code				26HSS132			Course Type				Ability Enhancement Course					
Teaching Hours/Week (L: T: P: S)				1:0:0:0			Credits				1					
Total Teaching Hours				15+0+0			CIE + SEE Marks				60+00					
Teaching Department: Centre For Flourish																
Course Objectives:																
1.		The course would support students to identify their strengths and work on the areas of improvement														
2.		The course will enable students to self reflect on their life goals and plan their action plan to attain them														
Unit - I																
SELF AWARENESS EXERCISES															7	
1. Ice breaker: Who am I? -to reflect on covert and overt attributes																
2. Tree of Life exercise- to reflect on their life wholistically																
3. Johari Window: -to identify and map interests, capacities and competencies; Reflections on personal strengths and identifying areas of improvement.																
Unit – II																
BEING A BETTER VERSION OF MYSELF															8	
1. Goal setting exercises: Short term and long term personal and professional goal setting; Reviewing the set goals and identifying immediate actionable steps to be taken.																
2. Identifying sources of distractions: action plan to manage and self-monitor																
3. Time management strategies: Dealing with procrastination; reducing screentime																
Course Outcomes: At the end of the course student will be able to																
1.		The student will be able to reflect on their strengths and capacities														
2.		Identify their areas of improvement														
3.		Set and submit short term and longterm professional and personal goals														
4.		Identify and manage their distractions														
	Program Outcomes→			1	2	3	4	5	6	7	8	9	10	11	PSO↓	
	↓ Course Outcomes														1	2
	26HSS132.1								1					3		
	26HSS132.2								1					3		
	26HSS132.3					2								3		
	26HSS132.4					2								3		
1: Low, 2: Medium, 3: High																

REFERENCE MATERIALS:

1.	Eurich, T. (2017). Insight: why we're not as self-aware as we think, and how seeing ourselves clearly helps us succeed at work and in life. New York, Crown Business.
2.	Seligman, M. E. P. (2011). Flourish: A visionary new understanding of happiness and well-being. Free Press.
3.	American Psychological Association. The Psychology of Procrastination: Why People Put Off Important Tasks Until the Last Minute. 2010.
4.	A psychological perspective: https://www.verywellmind.com/what-is-selfawareness-2795023#toc-how-does-self-awarenessdevelop
5.	Procrastination https://jamesclear.com/procrastination

Pathways to Success																		
Course Code			26HSS133			Course Type			Ability Enhancement Course									
Teaching Hours/Week (L: T: P: S)			1:0:0:0			Credits			1									
Total Teaching Hours			15+0+0			CIE + SEE Marks			60+00									
Teaching Department: Centre For Flourish																		
Course Objectives:																		
1.		The course would enable students to reflect and identify factors that are the building blocks to success																
2.		The course would provide an opportunity to students to draw up an action plan and practice good habits.																
Unit - I																		
NON-COGNITIVE FACTORS FOR SUCCESS															7			
1. Case studies: What habits are necessary to attain and sustain success?																		
2. Growth mindset or Fixed mindset? Assess yourself-analyze your self-talk.																		
3. Self-reflection: Review personal habits-action plan to develop and practice good habits																		
Unit – II																		
BEING A BETTER VERSION OF MYSELF															8			
1. Case studies: People who failed to succeed																		
2. Social Lab: Talk to people- life lessons- grit and resilience.																		
3. Self-reflection: How do I deal with setbacks and failures? - Do’s and Don’ts																		
4. Mindfulness practice																		
Course Outcomes: At the end of the course student will be able to																		
1.		Identify the non cognitive factors leading to success																
2.		Recognize failure as pathways to success																
3.		Recognise the makers of growth mindset in oneself																
4.		Practice techniques to build good habits																
		Program Outcomes→			1	2	3	4	5	6	7	8	9	1	1		PSO↓	
		↓ Course Outcomes												0	1		1	2
		26HSS133.1					2				1				3			
		26HSS133.2								2					3			
		26HSS133.3					1								3			
		26HSS133.4								2	1				3			
1: Low, 2: Medium, 3: High																		

REFERENCE MATERIALS:	
1.	Duckworth, A. (2016). Grit: The power of passion and perseverance. Scribner/Simon & Schuster.
2.	Khera, S. (2014) You Can Win: A Step-by-Step Tool for Top Achievers. A&C Black.
3.	TED talk by Angela Duckworth on grit-- https://www.youtube.com/watch?v=H14bBuluwB8&t=111s
4.	5 Practices of Highly Resilient People with Dr. Taryn Marie Stejskal: https://www.youtube.com/watch?v=gIYHQAeOy-w
5.	Learning from failure https://hbr.org/2011/04/strategies-for-learning-fromfailure
6.	Carol Dweck on Growth Mindset https://www.youtube.com/watch?v=hiiEeMN7vbQ